

Week 5: Multinational Firms & Capital Budgeting

FINA3020: International Finance

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Part I: Multinational Firms & Valuation

- **Foundations:** Trade costs, trade finance, and foreign direct investment (FDI)
- **Multinationals:** Productivity, market entry, and capital structure
- **Valuation:** International capital budgeting
- **Applications:** Taxes, blocked funds, transfer pricing, and governance
- **Game Round 5:** Foreign Project Committee

Part II: Midterm Exam Review

- FX Quotes & Triangular Arbitrage
- Parity Conditions (CIP, UIP, PPP)
- Carry Trade Mechanics & Regimes
- Exposure Measurement & Hedging

- A firm can serve foreign customers by exporting, licensing to an independent producer, or controlling a foreign affiliate. Each choice changes costs, risks, and financing needs.
- **Trade theory:** Which countries and firms produce which goods? Trade costs and comparative advantage determine where production is competitive.
- **Trade finance:** Who funds production and shipment, and who bears the risk of non-payment?
- **FDI and corporate finance:** When is foreign ownership worth its fixed cost, and how much value reaches the parent?

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Labor hours per unit	Textiles	Machines
Home	2	4
Foreign	6	6

Home is more productive in both goods, but its opportunity cost of one textile is only $1/2$ machine. Foreign's cost is one machine.

- Home has a comparative advantage in textiles. Foreign has a comparative advantage in machines, despite its absolute disadvantage.
- At a trade price of 0.75 machine per textile, Home obtains more machines per textile than at home, and Foreign pays fewer machines per textile than at home.
- This benchmark assumes competitive markets and no distortions. Trade also has distributional effects that can make some people worse off.

[Trade benchmark →](#)[Graph: exports good 1 →](#)[Graph: exports good 2 →](#)

A useful empirical benchmark for exports from country i to country n is

$$X_{ni} \approx A \frac{Y_i Y_n}{D_{ni}^\gamma}, \quad \gamma > 0.$$

- Larger economies buy and sell more. Distance proxies for shipping time and other frictions, including information and coordination costs.
- Tariffs, customs delays, language, and contract enforcement also affect delivered cost. Geographic distance is one metric. You can use a broader notion of distance instead
- For an MNC, a large but expensive-to-reach market may justify local production (horizontal FDI). Vertical FDI still needs affordable cross-border shipment of inputs.

[Ricardian gravity →](#)[Size evidence →](#)[Distance evidence →](#)[Regression interpretation →](#)[E&K →](#)[Conditional expectation →](#)[OLS geometry →](#)[Example →](#)

- Cash flow misalignment: Exporter pays wages and suppliers before the buyer pays the invoice. Production, transit, customs, and credit terms affect this funding gap.
- **Trade credit:** the supplier lets the buyer pay later. **Trade finance:** loans, guarantees, receivables finance, and related services that support trade.
- Collateral and reputation affect credit terms. Parent funding and export-credit agencies can help overcome financing constraints.

Illustrative funding cost

Funding USD 1M for 90 days at 8% simple annual interest costs
 $1,000,000(0.08)(90/360) = \text{USD } 20,000$, before fees or default losses.

[Invoicing and finance →](#)[Trade-finance geography →](#)[Trade credit diagram →](#)[Example →](#)

Method	Payment mechanism	Main exposure
Cash in advance	Buyer pays before shipment	Buyer bears delivery and performance risk
Letter of credit	Bank pays against complying documents	Documentary conditions, bank and country risk
Documentary collection	Banks release documents under agreed instructions	Seller bears buyer default risk
Open account	Goods ship before the invoice is paid	Seller extends credit and bears buyer default risk

Buyer bargaining power, trust, enforceability, order size, and banks' fees affect the choice of trade settlement.

Source: International Trade Administration, Trade Finance Guide.

[Open account steps →](#)[Cash in advance steps →](#)[Original risk ladder →](#)[Example →](#)

- **Documents against payment (D/P):** the collecting bank releases specified documents when the buyer pays. The seller can still face refusal or delay.
- **Documents against acceptance (D/A):** documents are released when the buyer accepts a promise to pay at a later date. Credit risk remains until payment.
- **Letter of credit:** the issuing bank undertakes to pay against a complying presentation. A confirming bank can add its own payment undertaking.

[Documentary collection steps →](#)[Letter of credit steps →](#)[Example →](#)

Instrument	What the exporter obtains	Key distinction
Factoring	Cash against short-term receivables, often with collection services	Recourse terms determine who bears covered default losses
Forfaiting	Cash from selling medium- or long-term export receivables	Usually without recourse, often with bank-backed debt claims
TCI	Protection against specified non-payment risks	Covers specified losses

Fees, exclusions, deductibles, and documentation affect what triggers recourse.

Source: ITA, Export Credit Insurance and Trade Finance Guide.

[Financing mechanisms →](#)[Recourse and insurance →](#)[Recourse comparison →](#)[Insurance diagram →](#)[Example →](#)

Transaction	Plausible starting point	Consider
Ghanaian retailer buys USD 5M of phones and parts from a large Vietnamese producer new to Ghana	Letter of credit	Bank fees, inspection, and an unfamiliar export destination
Brazilian boutique buys USD 50k of branded French perfume	Cash in advance	Seller reputation, delivery protection, and bargaining power
Hong Kong coffee chain buys USD 50k of Ethiopian beans	Documentary collection	Relationship history, country risk, title documents, and quality checks

Discussion: How would a long trading relationship change each answer? What changes if the exporter is a subsidiary of the buyer?

A **multinational corporation (MNC)** controls value-creating activities in more than one country through subsidiaries, branches, or affiliates operating across legal, tax, and currency jurisdictions.

- **Corporate Structure:** Headquartered in a home country; operates foreign subsidiaries subject to local laws, foreign currencies, and host-government regulations.
- **Internal Capital Markets:** MNCs can reallocate capital through intra-company loans, royalties, dividends, and equity injections, mitigating some local financing frictions while remaining subject to tax, legal, and capital-control constraints.
- **Empirical Facts:** Exporters and MNCs have systematically higher productivity, larger capital intensity, and higher R&D spending than purely domestic firms.

[View MNC Empirical Facts & Productivity Selection →](#)[Productivity evidence 1 →](#)[Productivity evidence 2 →](#)[Example →](#)

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- Multinational production is concentrated in relatively productive firms and is important in industries with substantial capital or R&D costs.
- The historical literature documents substantial two-way activity among developed economies. Cross-border acquisitions and greenfield projects serve different purposes.
- Headquarters often specialize in research, finance, and coordination. Affiliates may sell locally, supply the parent, or export to third countries.
- Export concentration, the share of sales through affiliates, and related-party trade have different denominators with different interpretations.

Source: Antràs and Yeaple (2014).

[Historical concentration →](#)

[Affiliate activity →](#)

[Example →](#)

- FDI is a cross-border investment relationship with at least 10% of voting power. It includes equity, reinvested earnings, and qualifying intercompany debt.
- Horizontal FDI serves a foreign market with local production. Vertical FDI places different production stages in different countries.
- Greenfield investment creates operations. An acquisition transfers ownership of existing operations. Buying an existing business is distinct from building new productive capital.
- Sometimes you see “affiliate” and “subsidiary” used interchangeably. I think of “subsidiary” as more specific: controlled, typically through a majority of voting rights.

Source: IMF, BPM6, chapters 6 and 11.

[Related-party trade →](#)

[Example →](#)

- An FDI flow measures a cross-border financing relationship. An FDI position measures its stock. Affiliate production, employment, and sales measure operating activity.
- Building a new factory creates productive capacity. Acquiring an existing firm changes ownership without itself adding the purchase price to GDP.
- FDI includes reinvested earnings. Affiliates can also borrow locally, so their total investment can exceed inward FDI.
- Local affiliate sales are distinct from exports from the parent country. Intra-group shipments are part of cross-country trade, even though buyer and seller share an owner.

[FDI accounting →](#)[Trade and FDI evidence →](#)[Example →](#)

- Exporting retains home production but incurs shipping, tariffs, and customs costs. Local production incurs setup and management costs but can avoid repeated costs on final goods.
- Outsourcing purchases from an independent supplier. Ownership offers control over quality and intellectual property, at the cost of management and regulatory responsibilities.
- Large host markets and high trade costs can favor horizontal FDI. Differences in input costs and production capabilities can favor vertical FDI.
- FDI can acquire brands, know-how, and distribution. Examples include Lenovo's acquisition of IBM's PC business and KFC's adaptation to the Chinese market. Franchising and licensing are alternatives to ownership.

Example →

	Independent supplier	Owned operation
Produce at home	Domestic outsourcing	In-house production for export
Produce abroad	Offshore outsourcing or licensing	Foreign affiliate (FDI)

- Location depends on delivered cost, market access, skills, and scale. Ownership depends on contracting, quality control, IP protection, and coordination.
- Standardized textile production is often suitable for outsourcing. Difficult-to-contract knowledge or quality can favor ownership.
- Horizontal FDI duplicates production near customers. Vertical FDI divides production stages across countries and can increase intermediate-goods trade.

[Example →](#)

- **Scale and input costs:** fixed costs make average cost fall with output. Compare unit costs across locations, including productivity, shipping, and coordination.
- **Capabilities:** buying a local firm can provide technology, brands, or distribution. Lenovo's acquisition of IBM's PC business illustrates this motive.
- **Local customers:** KFC's China expansion illustrates local adaptation. Nestlé's China product adaptations illustrate the same motive. Ownership, franchising, and licensing offer different control rights.
- **Demand diversification:** imperfectly correlated country demand can stabilize group cash flow. Common global shocks and exchange rates can offset the benefit.

Example →

Suppose a firm can serve the same demand q at price p through either exporting or a local affiliate:

$$\pi_X = (p - c_X)q - F_X, \quad \pi_I = (p - c_I)q - F_I.$$

If $F_I > F_X$ and $c_X > c_I$, local production is more profitable when

$$q > q^* = \frac{F_I - F_X}{c_X - c_I}.$$

Example: an affiliate adds USD 8M of fixed cost but saves USD 20 per unit. It needs more than 400,000 units to recover the extra fixed cost.

- A higher tariff can lower the break-even quantity. Cheaper shipping can raise it.
- This is a one-period illustration with identical prices and demand. The surgical-robot case adds construction time, growth, and discounting.

[Derivation step by step →](#)[Numerical example →](#)

A single producer

- With downward-sloping demand, selling more lowers the price. Profit maximization sets marginal revenue equal to marginal cost.
- Fixed costs make average cost decline with output. A markup over marginal cost can be needed just to cover those costs.

[Monopoly pricing →](#)[Entry equilibrium →](#)[Market integration →](#)[Example →](#)

Entry and market integration

- Profitable entry attracts competitors, lowering prices and sales per incumbent.
- A larger integrated market spreads fixed costs over more customers, supporting more varieties and lower prices in the benchmark.

Productivity	Typical activity	Economic reason
Highest	FDI	Large sales cover high affiliate setup costs
Intermediate	Exporting	Sales cover export entry costs and variable trade costs
Lower	Domestic only	Foreign-market profits fall short of entry costs
Lowest	Exit	Operating profits fall short of required costs

This ordering depends on costs and technology. Firms can combine activities. Higher observed MNC productivity can reflect selection into FDI.

[Selection details →](#)[Example →](#)

- An affiliate may gain access to parent funding and better financial management. Domestic suppliers may gain customers and access to working capital.
- Local borrowing by affiliates may crowd out other firms when domestic credit supply is constrained. It can also deepen markets and support new suppliers.
- FDI's effect on growth depends partly on local firms' ability to use imported technology and management practices.
- The Alfaro–Chauvin reading emphasizes that financing conditions help explain both the location and the effects of FDI.

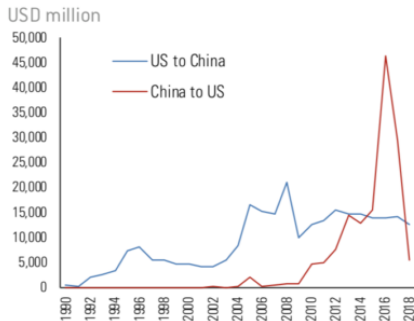
Capital flows and institutions →

Example →

- The 1979 equity joint-venture law established a legal route for foreign investment. Permitted foreign ownership varied by sector and period.
- Joint ventures can combine foreign capital and technology with local knowledge and labor to accelerate local productivity growth. There may be governance and intellectual-property concerns.
- China removed foreign ownership caps in new-energy vehicle manufacturing in 2018. Tesla's Shanghai factory is an example of wholly foreign-owned local production.
- The Foreign Investment Law took effect in 2020. Market access restrictions are separate from the general legal framework.

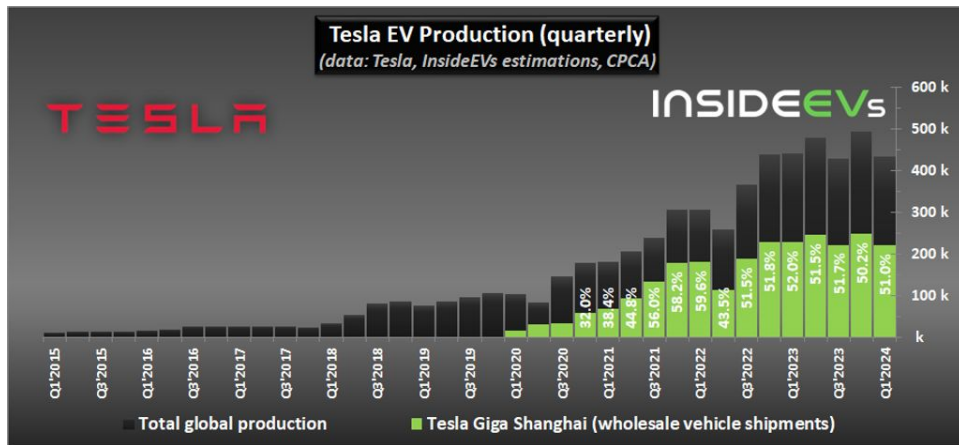
Example →

Figure ES-1: Annual Value of FDI Transactions between the US and China, 1990-2018*



Source: Rhodium Group. *See Appendix in the full report for data description.

Historical US–China FDI transactions. Estimating the effect of joint-venture rules requires accounting for other policies and market conditions.



Historical exhibit. Data period, coverage, and units are as labeled.

Hypothetical US firm, all amounts in USD millions. Unit price is 1.50. First-year sales are 500 units, growing 10% annually over a ten-year patent life.

Choice	Upfront cost	Unit cost / horizon
Export	0	1.00 production + 0.10 tariff + 0.05 shipping
Own factory	800	0.60, after one year of construction
Outsource	100	0.70, five sales years before IP leakage

- The tariff is assumed to apply to the USD 1M production value.
- Pay setup costs at date 0 and collect operating cash flow at each year end. Continue exporting during the factory's first year. Ignore tax and terminal salvage.
- Compare discounted operating profits using the same timing and price assumptions in all three choices.

All values are USD millions. Let k be the annual required return and $\beta = 1/(1+k)$ the one-year discount factor. Sales in year t are $q_t = 500(1.1)^{t-1}$. Thus $q_t\beta^t = 500\beta(1.1\beta)^{t-1}$.

Export Margin per robot is $1.50 - 1.15 = 0.35$, giving $500(0.35) = 175$ in Year 1. Export in all ten years, with no setup cost:

$$V_E = 175\beta \sum_{j=0}^9 (1.1\beta)^j.$$

Own factory Pay 800 now and export in Year 1 during construction. From Year 2, the margin is $1.50 - 0.60 = 0.90$. The coefficient is $500(0.90) = 450$, with growth beginning at $j = 1$:

$$V_F = -800 + 175\beta + 450\beta \sum_{j=1}^9 (1.1\beta)^j.$$

Outsource Pay 100 now. The margin is $1.50 - 0.70 = 0.80$, giving $500(0.80) = 400$. IP leakage ends sales after Year 5:

$$V_O = -100 + 400\beta \sum_{j=0}^4 (1.1\beta)^j.$$

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Discount factor	Export	Own factory	Outsource
$\beta = 0.9$	1,505.98	2,825.03	1,664.36
$\beta = 0.7$	493.59	276.72	787.87

- With a longer effective horizon, own production recovers its large setup cost. Heavy discounting favors the smaller upfront commitment to outsourcing (unit cost 0.70)

$$V_E = 175\beta \sum_{j=0}^9 (1.1\beta)^j.$$

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FDI Trend: China vs Other Emerging Markets (EM)

Figure 4. EM ex-China: Non-Resident Flows
(4Q, % of GDP)

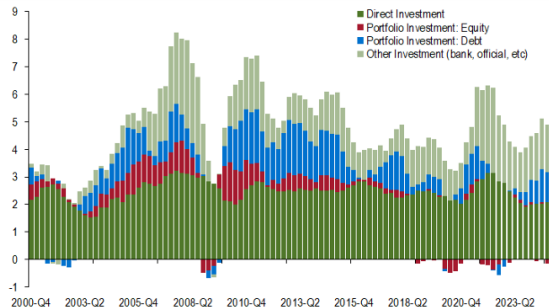
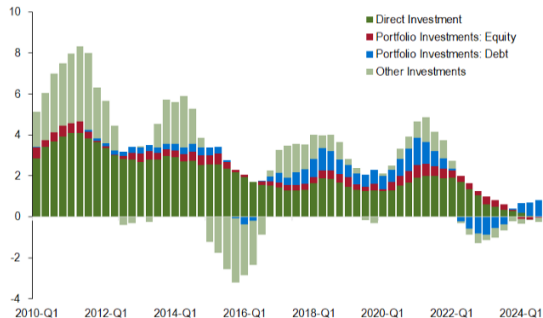


Figure 6. China Non-Resident BOP Flows
(4Q % to GDP)



Historical exhibit. Data period, coverage, and units are as labeled.

Source: IMF.

FDI Trend: China vs Other Emerging Markets (EM)

Figure 9. EM BOP Resident Outflows
(4Q % to GDP)

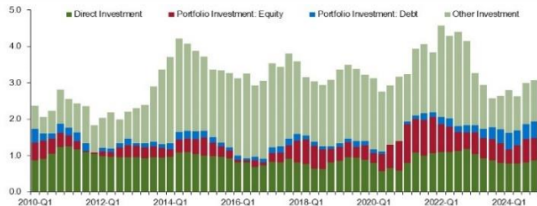
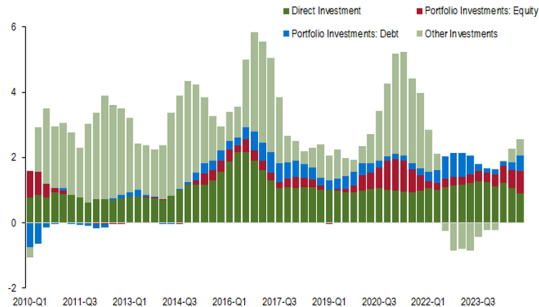


Figure 11: China Resident Flows
(4Q % to GDP)



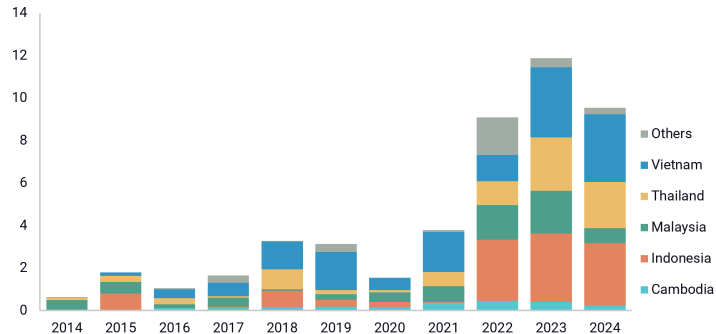
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Announced Chinese Greenfield Manufacturing FDI in ASEAN

Value of major announced greenfield Chinese manufacturing FDI in ASEAN by economy
(excluding extractive industries)

USD billion

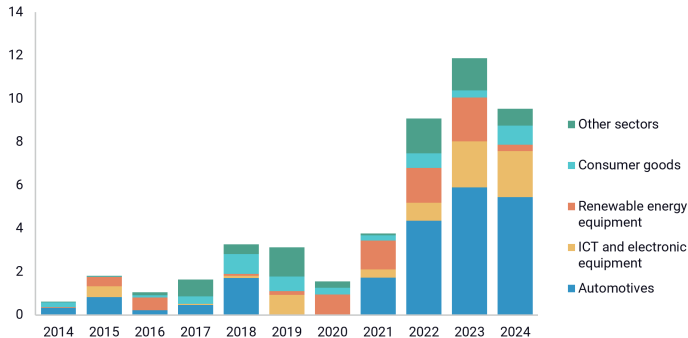


Source: Rhodium Group China Cross-Border Monitor. Note: Major transactions include investments above \$5 million only. Extractive industries include investments in the energy and minerals sectors.

Announced Chinese greenfield manufacturing FDI in ASEAN, by destination. Announcements differ from realized flows.

Announced Chinese Greenfield Manufacturing FDI in ASEAN

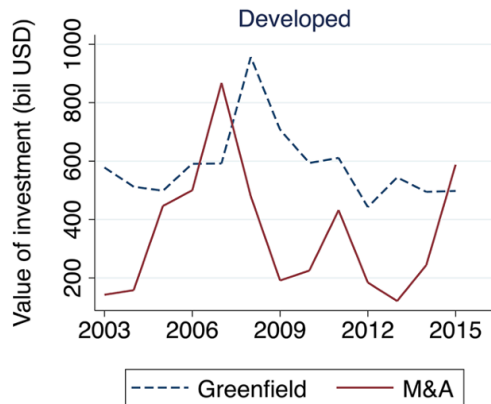
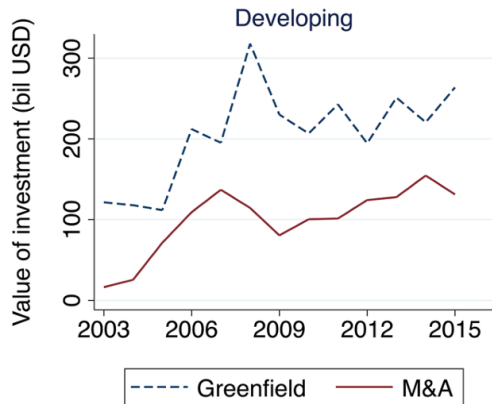
Value of major announced greenfield Chinese manufacturing FDI in ASEAN, by sector
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FDI Trend: Greenfield vs M&A FDI



Historical exhibit. Data period, coverage, and units are as labeled.

Source: Chan & Zheng (2022) using data from UNCTAD.

How does operating across borders affect an MNC's optimal debt-to-equity ratio?

- **Modigliani-Miller (MM) Benchmark:** In frictionless global capital markets without taxes or default costs, firm value V and cost of capital are independent of debt ratio and currency choice.
- Why MM Fails in International Markets:
 - ① Tax Asymmetries: Interest deductions vary across host tax regimes.
 - ② Bankruptcy Frictions: Foreign subsidiary defaults involve cross-border legal complexity.
 - ③ Currency & Sovereign Risks: FX swings alter local debt service capacity.

[View MM Arbitrage Proof & International Violations →](#)

[Example →](#)

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[Example →](#)

A hurdle rate should match the risk and currency of the project cash flows. With market values E, D , the standard after-tax WACC is

$$\text{WACC} = \frac{E}{D + E} k_E + \frac{D_{\text{bonds}} + D_{\text{loans}}}{D + E} k_D (1 - \tau).$$

Source	Market value	Cost
Equity	USD 1.0B	8%
Bonds	USD 0.7B	6% after tax
Loans	USD 0.3B	5% after tax
Total $D + E$	USD 2.0B	

$$\text{WACC} = \frac{1.0}{2.0} (8\%) + \frac{0.7}{2.0} (6\%) + \frac{0.3}{2.0} (5\%) = 6.85\%.$$

- Each color links a table entry to its place in the formula. Gray is the total market value $D + E$.
- Use after-tax debt costs directly in WACC. Apply firm-wide WACC to projects with comparable risk and financing.

Assume no taxes, distress costs, or trading frictions. Two firms own the same perpetual assets. Annual asset cash flow is 150 or 50 with equal probability. The asset required return is 10%.

Claim	High state	Low state	Value today
All assets	150	50	1,000
Perpetual debt at 5%	10	10	200
Levered equity	140	40	800

$$k_E = \frac{0.5(140) + 0.5(40)}{800} = 11.25\%, \quad \text{WACC} = 0.8(11.25\%) + 0.2(5\%) = 10\%.$$

- Debt reallocates cash flows and raises equity risk. In this debt-financed payout example, total asset value stays 1,000 while equity falls to 800.

Example →

Leverage and Unleveraging

Replication	Initial outlay	Annual high / low payoff
Buy unlevered firm, borrow 200	$1,000 - 200 = 800$	$150 - 10 / 50 - 10$
Buy levered equity and its debt	$800 + 200 = 1,000$	$140 + 10 / 40 + 10$

- The first portfolio reproduces levered equity. The second reproduces unlevered assets, including any default states if the actual firm debt is purchased.
- If levered total value exceeds unlevered value, sell the expensive claims and buy their cheaper replication. Reverse the trade when levered total value is lower.
- Replication requires matched cash flows and financing access. Taxes, borrowing spreads, capital controls, and legal differences prevent the simple arbitrage for many MNCs.

Overpriced leverage →

Underpriced leverage →

$$V_L = V_U + PV(\text{usable tax shields}) - PV(\text{distress and agency costs}).$$

For permanent debt D with fully usable interest deductions, the textbook tax-shield value is τD . This requires strong assumptions about debt and tax-shield risk.

Illustrative debt	0	200	400	600
Total firm value	700	720	735	730

- Borrowing initially adds value, but expected legal costs, lost customers, and conflicts between creditors and shareholders eventually outweigh extra tax benefits.
- MNCs choose the amount, currency, and entity that borrows. Local tax shields, bankruptcy rules, guarantees, and cash-flow matching jointly determine the choice.

Debt location and tax →

Example →

- Managers know more than outside investors about project value. Investors may discount newly issued securities when they suspect adverse selection.
- Internal cash avoids selling claims to outsiders. Relatively safe debt is less sensitive to private information than equity, motivating the order: retained earnings, debt, then equity.
- Equity issuance can be interpreted as a signal of overvaluation in the model. It can also finance good projects when debt capacity is exhausted.
- Under the pecking order, the debt ratio reflects past funding needs and retained earnings.
- For a Brazilian affiliate, compare parent retained earnings, local BRL debt, and equity. Consider currency risk, information, and regulatory constraints.

Example →

International Trade-Off Theory

- Balances tax shield gains ($\tau_c \cdot D$) against distress costs.
- Higher local tax rates can increase interest tax shields. Deduction limits, withholding, transfer-pricing rules, and losses can reduce the benefit.
- Matches local currency revenues with local currency debt service to reduce FX exposure.

International Pecking Order

- Financing order driven by cross-border information asymmetry:
 - ① Internal cash flows (headquarters or subsidiary retained earnings).
 - ② Debt from the parent, local banks, or other lenders; access and information determine the source.
 - ③ External equity, which can be costly when investors know less than managers.

Summary

Choose the borrowing location by comparing tax treatment, legal recourse, financing access, and currency matching across jurisdictions.

- Trade-off theory balances the usable interest tax shield against distress and agency costs. The value of a deduction usually rises with the applicable marginal tax rate.
- Pecking-order theory predicts internal funding, then debt, then equity when information problems make external claims costly. The predicted order depends on information costs and financing access.
- A parent can move retained earnings to a constrained affiliate, provide a guarantee, or arrange an intercompany loan. Currency matching may justify more expensive local funding.
- Example: for an affiliate in Brazil, compare parent funds, BRL bank debt, and equity. Consider information costs, exchange-rate exposure, minority rights, and the parent's remaining debt capacity.

Example →

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Example →

- Capital controls, investor mandates, disclosure, and political risk can prevent investors from pricing otherwise similar claims in a single integrated market.
- Cross-listing can broaden an investor base. A parent guarantee can improve affiliate access to credit, while exposing the parent to contingent liabilities.
- Local-currency borrowing may have a higher nominal rate but reduce the mismatch between revenue and debt service. Compare cash flows in a common currency and account for hedging.
- Use a discount rate for the project's risk and cash-flow currency. Project risk determines the required return, while group financing affects access to funds.

Sensitivity Example: a USD 10M perpetuity is worth USD 100M at 10%, but USD 76.92M at 13%.

Example →

Evaluating foreign capital expenditure (CAPEX) requires converting cash flows and discount rates across currencies consistently. Two valuation methods:

- **Home Currency Approach (Parent Perspective):** Forecast foreign cash flows CF_t^* , translate each date- t flow using a consistently risk-adjusted exchange-rate forecast $E_0[S_t]$, and discount at a home-currency required return k_h appropriate for those cash flows.
- **Foreign Currency Approach (Subsidiary Perspective):** Discount foreign cash flows CF_t^* at Foreign WACC k_f , convert resulting foreign NPV to home currency at current spot rate S_0 .

The two methods agree only when project risk, FX risk, cash-flow forecasts, and currency-specific discount rates are priced consistently. UIP and relative PPP describe exchange-rate relations. Project WACCs also reflect operating and financial risk.

[View Proof of Valuation Equivalence →](#)

[Example →](#)

Value cash flows and assets available to the parent after tax, remittance, financing, and capital-control constraints.

Why do subsidiary & parent cash flows differ?:

- ① **Taxation Frictions:** Foreign corporate income tax vs. home tax credit ceilings.
- ② **Remittance Restrictions:** Host government limits on dividend repatriations (blocked funds).
- ③ **Intra-Company Overhead:** Licensing fees, management charges, and transfer prices paid to parent.
- ④ **Exchange Rate Risk:** Spot rate movements between cash generation and dividend remittance.

[View Step-by-Step 3-Year Project Example →](#)

[Example →](#)

Host governments generally tax local subsidiary income. Home-country treatment varies: dividends may be exempt, credited, deferred, or subject to minimum-tax and anti-deferral rules.

- **Foreign Tax Credit (FTC):** In a credit system, qualifying foreign taxes may offset home tax on foreign-source income, subject to jurisdictional or income-basket limitations and domestic-law eligibility rules.

Usable FTC $\leq$ Home tax attributable to the relevant foreign-source income

- Two tax outcomes:
 - In a simplified worldwide credit system, a lower qualifying foreign tax can leave residual home tax.
 - If qualifying foreign tax exceeds the limitation, current home tax may be zero. Use of excess credits depends on domestic carryover rules.

[View Detailed Tax Credit & Remittance Withholding Calculation →](#)

Blocked Funds (Remittance Limits)

- Host country capital controls restrict dividend transfers abroad.
- Unrepatriated cash may be reinvested locally, held in liquid assets, or released later; the attainable after-tax return matters.
- A restriction reduces value when the available local uses earn less than the parent could earn elsewhere, after adjusting for risk.

[Tax credits and blocked funds →](#)[Example →](#)

Transfer Pricing Strategy

- Setting internal prices for components, IP, or management fees.
- Transfer prices allocate income and customs value across entities; they can affect tax and remittance outcomes in more than one jurisdiction.
- Prices must follow applicable arm's-length, documentation, substance, customs, and anti-avoidance rules.

- A transfer price assigns revenue and cost to units within a group. It affects reported divisional profit, tax, performance targets, and internal investment decisions.
- The arm's-length principle asks what independent parties would charge in comparable circumstances. Applicable tax rules constrain the internal prices firms can use.
- Comparable transactions are harder to identify for unique intellectual property. Documentation, audits, and advance pricing agreements can reduce disputes.
- A manager rewarded on divisional profit may resist a group-optimal transfer price. Consolidated value requires including cash flows and taxes in every affected entity.

Source: OECD Transfer Pricing Guidelines (2022).

Example →

Hypothetical tax rates: German subsidiary 30%, US parent 20%. The subsidiary makes an item for EUR 80 and sells it to the parent, which resells for EUR 200.

Transfer price	German profit	US profit	Combined tax
EUR 100	20	100	$0.30(20) + 0.20(100) = 26$
EUR 140	60	60	$0.30(60) + 0.20(60) = 30$

- Raising the price moves EUR 40 of profit toward higher-tax Germany and increases tax by EUR 4.
- If instead the subsidiary buys from the parent, raising the transfer price moves profit in the opposite direction.
- These arithmetic scenarios ignore withholding, residual tax, customs, and enforcement. Actual prices must satisfy applicable rules.

$$PV_t = \frac{\left(\overbrace{\text{Rev} - \text{Costs} - \text{Dep}}^{\text{EBIT}} - \overbrace{\tau \times \text{EBIT}}^{\text{Corp. tax}} + \text{Dep} - \text{CAPEX}_t - \Delta\text{NWC}_t \right) \times (1 - \text{WHT}) \times \text{FX}_t}{(1 + r)^t}.$$

$$\text{FX}_t = \begin{cases} E_0[S_t] & \text{unhedged: convert at the expected spot rate} \\ F_{0,t} & \text{hedged: sell the euro amount forward at the forward rate} \end{cases} \quad (\text{USD per 1 EUR})$$

- Use incremental cash flows. Depreciation is noncash: deduct it to compute tax, then add it back. The parentheses give free cash flow to the subsidiary.
- Withholding tax applies to the cash remitted, and the FX rate converts it to USD. A forward fixes $F_{0,t}$ but not the euro amount, so hedge the amount that you will have.

Example →

Year 1 operating inputs (EUR millions): Revenue = 12.0, cash operating costs = 6.0, depreciation = 2.0, local corporate tax rate = 20%, and no incremental CAPEX or working-capital investment in Year 1.

Cash Flow Step	EUR Millions
Revenue	12.00
Less: Cash operating costs	(6.00)
Less: Depreciation	(2.00)
Earnings before tax	4.00
Less: Local corporate tax ($4.00 \times 20\%$)	(0.80)
Net operating profit after tax	3.20
Add back: Depreciation (non-cash)	2.00
After-tax operating cash flow	5.20

Example 2: Convert and Discount Parent Cash Flows

[Core Content]

Project assumptions: Initial CAPEX = EUR 10.0M; $S_0 = 1.10$ USD per 1 EUR; subsidiary free cash flows = EUR 5.2M, 6.4M, and 12.6M; all cash is remittable with no additional parent tax; expected future spot rates = 1.12, 1.14, and 1.16 USD per 1 EUR; home required return = 10%.

Year	EUR Cash Flow	Spot Rate	USD Parent Cash Flow	Present Value at 10%
0	(10.000)	1.10	(11.000)	(11.000)
1	5.200	1.12	5.824	5.295
2	6.400	1.14	7.296	6.030
3	12.600	1.16	14.616	10.981

$$PV_t = \frac{CF_t^{\text{EUR}} \times S_t}{(1+r)^t}, \quad NPV_{\text{USD}} = -11.000 + 5.295 + 6.030 + 10.981 = +\$11.306\text{M}$$

Baseline Decision

The project creates value before new remittance restrictions: **accept**. The game asks whether that decision survives a policy shock.

Example 2b: Hedge the Parent Cash Flows with Forwards

[Core Content]

Same project as Example 2. The firm sells each year's expected euro cash flow forward at $F_{0,1} = 1.11$, $F_{0,2} = 1.12$, $F_{0,3} = 1.13$ USD per 1 EUR. Rates are USD per 1 EUR, so a lower spot rate means the USD has appreciated.

Year	EUR Cash Flow	Forward Rate	Hedged USD Cash Flow	Present Value at 10%
0	(10.000)	1.10	(11.000)	(11.000)
1	5.200	1.11	5.772	5.247
2	6.400	1.12	7.168	5.924
3	12.600	1.13	14.238	10.697

$$PV_t = \frac{CF_t^{\text{EUR}} \times F_{0,t}}{(1+r)^t}, \quad NPV_{\text{hedged}} = -11.000 + 5.247 + 5.924 + 10.697 = +\$10.868\text{M}.$$

- Hedging lowers expected NPV by $11.306 - 10.868 = \$0.437\text{M}$, because each forward rate is below the expected spot rate.
- The hedged NPV is the same at any future spot rate. If every future spot rate is 1.00 (USD appreciates), the unhedged NPV is $+\$8.483\text{M}$. If every future spot rate is 1.24 (USD depreciates), it is $+\$13.159\text{M}$.

Shock to Year 2 cash flow: EUR 6.4M faces a 20% dividend-remittance withholding tax and a one-year freeze. Blocked funds earn 3% locally and are released in Year 3; $E[S_3] = 1.16$ USD per 1 EUR and home WACC = 10%.

Baseline: immediate Year 2 remittance

$$PV_0 = \frac{6.4(1.14)}{1.10^2} = \$6.030\text{M}$$

Withholding plus one-year freeze

$$PV_0 = \frac{6.4(0.80)(1.03)(1.16)}{1.10^3} = \$4.596\text{M}$$

Withholding only (before any FTC)

$$PV_0 = \frac{6.4(0.80)(1.14)}{1.10^2} = \$4.824\text{M}$$

Value lost on the Year 2 cash flow

$$6.030 - 4.596 = \$1.434\text{M}$$

Comprehension Check

When withholding tax qualifies for a credit, total tax equals withholding tax plus home-country tax after usable foreign tax credits.

All operating amounts are EUR millions. Initial investment is EUR 30 at EUR/USD = 1.10. Depreciable basis is EUR 27, annual depreciation EUR 5, and terminal book value and salvage both EUR 12. The remaining initial EUR 3 is a nonrecoverable setup cost.

Year	Revenue	Cash cost	Taxable profit	After-tax cash
1	30	18	7	10.25
2	32	19	8	11.00
3	34	20	9	11.75 + 12 salvage

- Assume host corporate tax 25%, 10% withholding on all modeled distributions including terminal cash, no additional home tax, and full remittance.
- In reality, capital returns and dividends may receive different treatment.

[Year 1 calculation →](#)[Year 2 calculation →](#)[Year 3 calculation →](#)

Year	After withholding, EUR M	Converted USD M	PV at 10%, USD M
1	9.225	10.1475	9.2250
2	9.900	10.8900	9.0000
3	21.375	23.5125	17.6653

$$\text{NPV} = 9.225 + 9 + 17.6653 - 30(1.10) = \text{USD } 2.8903\text{M}.$$

- If initial FX stays 1.10 but every future conversion is 1.05, NPV becomes $35.8903(1.05/1.10) - 33 = \text{USD } 1.2590\text{M}$.
- If every future conversion is 1.15, NPV becomes USD 4.5217M. Revaluing the initial investment too is a different experiment.
- The sequence is operating cash flow, local tax, remittance restrictions, withholding and usable tax credits, currency conversion, then discounting.

Continue the three-year exercise. In Year 1 only, the subsidiary pays an additional EUR 4M to the parent for inputs. Assume no incremental real cost, a 20% parent tax on that markup, and no tax credit or other tax interaction.

Incremental effect	EUR M
Subsidiary tax saved: 25% of 4	+1.00
Subsidiary cash before withholding	-3.00
Reduced remittance after 10% withholding	-2.70
Parent markup after 20% tax	+3.20
Consolidated incremental parent cash	+0.50

$$\Delta NPV = 0.50(1.10)/1.10 = \text{USD } 0.50\text{M.}$$

- The subsidiary's remittance PV falls from 9.225 to 6.525, but that omits the parent's input-sale income. The consolidated calculation includes both.

Role: CFO Investment Committee evaluating a 3-year CAPEX expansion in Emerging Market X.

- **Stage 1:** Submit ID. Review the baseline: Initial outlay = EUR 10M; foreign cash flows = EUR 5.2M, 6.4M, and 12.6M; spot forecast = 1.12, 1.14, 1.16 USD per 1 EUR; home WACC = 10%.
- **Stage 2:** Compute baseline home currency NPV, unhedged and with a forward hedge ($F_{0,1}, F_{0,2}, F_{0,3} = 1.11, 1.12, 1.13$ USD per 1 EUR). Choose the FX policy and submit the initial Accept / Reject choice. The game has four rounds with new numbers, hint buttons, and a Big Text button for the screen.
- **Stage 3 [Exogenous Shock]:** Host government imposes a 20% withholding tax and 1-year freeze on Year 2 cash flows. Re-calculate parent NPV under alternative Foreign Tax Credit rules, with and without the hedge. The frozen cash flow arrives in Year 3, so re-hedge it at $F_{0,3}$.
- **Stage 4:** Debrief optimal decision vs common valuation errors.
- **Stage 5:** Individual Exit Reflection submission: *"How did usable foreign tax credits change the NPV impact of the withholding tax?"*



[Open Week 5 Game](#)

- Shareholders delegate decisions to managers. Distance, language, local regulation, and complex group structures make monitoring costly.
- Parent and affiliate managers can disagree over investment, transfer pricing, and retention of cash. Controlling shareholders can also conflict with minority owners.
- Boards oversee strategy and management. Independent directors, local expertise, internal controls, external audits, and enforceable disclosure work together.
- Cross-listing can broaden the investor base and impose additional disclosure. Governance benefits depend on the listing program, enforcement, and actual oversight.

[Financial statements →](#)[MNC examples →](#)[Company comparisons →](#)[Example →](#)

Tunneling, Disclosure, and Enforcement

- Tunneling transfers value from a firm to its controller through unfair related-party sales, loans, guarantees, or dilution. Ordinary arm's-length intragroup trade is different.
- The Samsung case in Bae, Kang, and Kim (2002) illustrates the risk of transfers favoring a controlling family. Enron illustrates concealed liabilities and misleading earnings through complex entities.
- The Renault–Nissan alliance illustrates conflicts among partners and public shareholders. Volkswagen and Siemens illustrate the importance of oversight and cross-border enforcement.
- Legal-origin research reports historical associations between institutions and finance. Assess current shareholder rights, enforcement, and firm governance when evaluating financing costs.

Example →

- **Shareholders and managers:** managers may expand the group, retain cash, or pursue private benefits even when shareholders would prefer a payout.
- **Headquarters and affiliate managers:** local managers know more about customers and costs. Headquarters must distinguish weak effort from adverse local conditions.
- **Controlling and minority shareholders:** a controller can benefit another affiliated company through an unfair price, loan, guarantee, or share issue.
- **Creditors and shareholders:** risky investment or asset transfers can reduce the value of existing debt. Guarantees can move exposure across entities.

Renault–Nissan →

Enron →

Samsung and tunneling →

- Boards oversee strategy, senior management, and major related-party transactions. Managers run daily operations.
- Distance, language, entity boundaries, and differences in disclosure complicate monitoring. Local knowledge and independent judgment are both valuable.
- Useful controls include consolidated reporting, independent audits, review of guarantees and related-party transactions, and clear escalation of compliance concerns.
- Board effectiveness depends on incentives, information, expertise, and enforcement.

Vale and monitoring →

Volkswagen oversight →

- Investor protection includes voting and disclosure rights, remedies for abusive transactions, and credible enforcement. Written rights and practical access to remedies can differ.
- Legal-origin research compares historical country patterns, such as those in the UK and France. Evaluate each firm's shareholder rights, enforcement, and oversight under current rules.
- Cross-listing places shares or depositary receipts in another market. An ADR represents an interest in foreign shares, and program types carry different reporting obligations.
- A broader investor base and stronger disclosure can lower financing frictions. Listing costs and compliance obligations also rise, and misconduct remains possible.

[Petrobras ADRs →](#)[Legal origins →](#)[Siemens enforcement →](#)

- **Entry:** compare exporting, independent contracting, and owned production using trade costs, scale, local demand, and control requirements.
- **Funding:** compare the amount, currency, and location of borrowing, internal funds, and equity. Financing frictions can change feasible investment.
- **Value:** calculate incremental operating cash, taxes, remittances, terminal cash, FX conversion, and discounting from the parent's perspective.
- **Governance:** establish who makes decisions, who bears risk, and who can transfer value across entities. Group accounts combine results across entities, while managers and owners can still have conflicting incentives.

Example →

Exam Logistics

- **Date / Time:** Week 6 (First 75 minutes of class).
- **Weight:** 25% of course total.
- **Scope:** Weeks 1, 2, 3, and 4.
- **Rules:** Closed-book, closed-notes. Simple non-programmable calculator allowed. No AI tools, laptops, or smartphones.

Exam Format

- **Section A:** Conceptual Multiple Choice / Short Questions (30%).
- **Section B:** Quantitative Problem Solving & Arbitrage Execution (50%).
- **Section C:** Policy & Risk Intuition Transfer (20%).

Core Concepts to Master

- Quotation convention: BASE/QUOTE (Base = 1 unit, Quote = price).
- Bid-Ask Rule: Dealer buys base at Bid, sells base at Ask. Customer trades on opposite side!
- Implied Cross-Rate calculation with bid-ask spreads.

[View Week 1 Sample Exam Problem & Solution →](#)

Triangular Arbitrage Steps

- ① Compare implied cross-rate with direct market quote.
- ② Trace 3-step currency loop (Clockwise vs. Counter-Clockwise).
- ③ Multiply execution rates (sell base at bid, buy base at ask). Profit if loop product > 1 .

Let S and F be home currency per foreign unit, with rates matched to horizon T .

- **CIP**: covered home and foreign investments have equal payoffs in the frictionless benchmark.

$$F^* = S \frac{1 + r_h T}{1 + r_f T}.$$

Executable arbitrage requires a discrepancy large enough to cover trading and financing costs.

- **UIP**: with zero expected excess currency return,

$$E_t[S_{t+T}] = S_t \frac{1 + r_h T}{1 + r_f T}.$$

Risk premia and expectation errors can produce observed excess returns.

- **PPP**: $S_{PPP} = P_h/P_f$. The real exchange rate is $q = SP_f/P_h$.

Carry Trade Strategy

- Borrow in low-yield funding currency (r_F low) and invest in high-yield target currency (r_H high), unhedged.
- If S is funding-currency units per target-currency unit, approximate expected excess return is $(r_H - r_F) + E[\Delta S/S]$.
- Positive historical average excess returns are sample-dependent and exposed to currency, crash, funding, and liquidity risk.

[View Week 3 Sample Exam Problem & Solution →](#)

Crash Risk & Regimes

- **Tail Risk:** High-yield currencies crash sharply during market stress ("up the stairs, down the elevator").
- **Exchange Rate Regimes:** Fixed/Pegged vs Crawling vs Floating; Policy Trilemma trade-offs.

Comparing the Four Hedging Strategies for a Foreign Receivable (CF^*):

Strategy	Payoff / Lock Formula	Key Operational Feature
1. Unhedged	$CF^* \cdot S_{t+1}$	Full upside participation, full downside risk
2. Forward Lock	$CF^* \cdot F_t$	Fixes the conversion rate for a certain amount/date; retains counterparty and settlement risk
3. Money Market	$CF^* \cdot \frac{1}{1+r_f} \cdot S_0 \cdot (1 + r_d)$	Idealized equivalent to a forward under CIP and matched funding rates
4. FX Option	$\max(CF^* K, CF^* S_{t+1})$ FV of premium	— Floor on downside, preserves appreciation upside

- ① **Quote Direction:** Always identify which currency is BASE and which is QUOTE before calculating.
- ② **Bid/Ask Side:** When selling base currency, use dealer's **BID**. When buying base currency, use dealer's **ASK**.
- ③ **Tenor Alignment:** Convert annual rates using the quoted instrument's day-count and compounding convention (e.g., Actual/360 where applicable).
- ④ **Cash Flow Timing:** Map out cash flow timing diagram before taking Present Value (PV).
- ⑤ **Economic Intuition:** Explain the economic mechanism behind your numerical answer in 1–2 clear sentences.

Example →

- MNC facts, market entry, productivity, and capital structure
- Capital budgeting, tax credits, annual calculations, and transfer prices
- Midterm practice solutions for Weeks 1–4
- Trade models, settlement mechanisms, and FDI accounting
- Governance cases and cross-border enforcement

Each idea slide links to an example or case study. Use the return button to resume the main deck.

[← Back to Main Slide](#)

- **Melitz Model Selection:** Foreign entry involves substantial fixed costs (establishing subsidiaries, regulatory compliance). Only the most productive firms earn sufficient operating profits to cover foreign entry costs.
- **Empirical Regularities:**
 - MNCs are generally larger, more productive, more capital intensive, and pay higher average wages than purely domestic firms, partly because more productive firms self-select into foreign activity.
 - Related-party trade is an important part of international trade, but its measured share varies by country, definition, and dataset.

Consider an MNC with unhedged operating cash flows $\tilde{Y}^*$ in foreign currency.

- **Unlevered Firm Value:** $V_U = \frac{E[\tilde{Y}^* \cdot \tilde{S}]}{1+k_u}$.
- **Levered Firm Value:** $V_L = E + D$.
- **Arbitrage Proof:** If $V_L > V_U$, an investor can sell levered firm shares, buy unlevered firm shares, and borrow on personal account (homemade leverage) to achieve the exact same payoff at a lower total cost.
- **Conclusion:** Financing choices can affect firm value through tax shields, market segmentation, or risk management.

Let CF_t^* be a foreign-currency cash flow and S_0 be home currency per foreign unit. Suppose the risk-adjusted discount rates and conversion factors satisfy

$$\hat{S}_t = S_0 \left(\frac{1 + k_h}{1 + k_f} \right)^t.$$

Then the home-currency calculation gives

$$\begin{aligned} \text{NPV}_h &= -S_0 CF_0^* + \sum_{t=1}^T \frac{CF_t^* \hat{S}_t}{(1 + k_h)^t} \\ &= S_0 \left[-CF_0^* + \sum_{t=1}^T \frac{CF_t^*}{(1 + k_f)^t} \right] = S_0 \text{NPV}_f. \end{aligned}$$

This equality requires consistent project discount rates and FX conversion. UIP relates interest rates and expected exchange rates; project WACCs also reflect operating and financial risk.

Project Data: Parent = US firm (USD). Foreign Sub = Eurozone (EUR). Initial CAPEX = EUR 10M ($S_0 = 1.10$ USD per 1 EUR). Home required return $k_h = 10\%$. Terminal recovery is EUR 5M after any disposal tax. No additional CAPEX or working-capital investment is assumed.

Line Item (EUR Millions)	Year 1	Year 2	Year 3
1. Operating Revenue	12.00	15.00	18.00
2. Operating Expenses	(6.00)	(7.50)	(9.00)
3. Depreciation	(2.00)	(2.00)	(2.00)
4. Earnings Before Tax (EBT)	4.00	5.50	7.00
5. Local Corporate Tax (20%)	(0.80)	(1.10)	(1.40)
6. Net Operating Profit After Tax	3.20	4.40	5.60
7. Add Back Depreciation	2.00	2.00	2.00
8. Salvage Value / Recovery	—	—	5.00
9. Free Cash Flow to Sub (EUR)	5.20	6.40	12.60
10. Expected Spot Rate (USD per EUR)	1.12	1.14	1.16
11. Repatriated Cash Flow to Parent (USD)	5.824	7.296	14.616

$$\text{Parent NPV} = -11.000 + \frac{5.824}{1.10} + \frac{7.296}{1.10^2} + \frac{14.616}{1.10^3} = +\$11.306\text{M}$$

Case 1: Simplified Foreign Tax Credit Ceiling Suppose a home jurisdiction uses a pure worldwide credit system. Foreign-source taxable income is EUR 10M, qualifying foreign tax is 25%, and the home tax rate is 21% (ignoring baskets, participation exemptions, minimum taxes, and expense allocation).

- Local Tax Paid = EUR 2.5M.
- Tentative Home Tax before credit = $\text{EUR } 10\text{M} \times 21\% = \text{EUR } 2.1\text{M}$.
- Usable FTC = $\min(2.5\text{M}, 2.1\text{M}) = \text{EUR } 2.1\text{M}$. Current residual home tax is zero. Use of the EUR 0.4M excess in other periods depends on domestic carryover rules.

Case 2: Blocked Funds Discounting Suppose Year 1 cash flow of EUR 5.2M is blocked by host government for 2 years and earns only 2% local bank interest, while parent cost of capital is 10%.

- Accumulated Cash in Year 3 = $\text{EUR } 5.2\text{M} \times (1.02)^2 = \text{EUR } 5.41\text{M}$.
- PV at Parent in Year 0 = $\frac{5.41\text{M} \cdot E_0[S_3]}{(1+k_h)^3}$, creating a significant value reduction vs immediate remittance.

Problem: You observe the following quotes from two dealers: Dealer A: EUR/USD = 1.0820 – 1.0825. Dealer B: GBP/USD = 1.2640 – 1.2648. Dealer C: GBP/EUR = 1.1650 – 1.1658. Is there a profitable triangular arbitrage? **Solution:**

- ① Calculate implied GBP/EUR cross-rate from Dealers A and B:

$$\text{Implied Bid}_{\text{GBP/EUR}} = \frac{\text{Bid}_{\text{GBP/USD}}}{\text{Ask}_{\text{EUR/USD}}} = \frac{1.2640}{1.0825} = 1.16766$$

$$\text{Implied Ask}_{\text{GBP/EUR}} = \frac{\text{Ask}_{\text{GBP/USD}}}{\text{Bid}_{\text{EUR/USD}}} = \frac{1.2648}{1.0820} = 1.168946$$

- ② Compare Implied Cross-Rate (1.1677 / 1.1690) with Dealer C quote (1.1650 / 1.1658).
- ③ Dealer C's Ask (1.1658) is LOWER than Implied Bid (1.1677)! Buy GBP from Dealer C at 1.1658 EUR, sell GBP for USD at Dealer B (1.2640), buy EUR with USD at Dealer A's ask (1.0825).
- ④ Profit per EUR 1M traded = EUR 1,000,000 $\times$ $\left(\frac{1}{1.1658} \times 1.2640 \div 1.0825\right) - 1,000,000 = +\text{EUR } 1,602$ (before fees and size limits).

Problem: Spot $S_0 = 7.8000$ HKD per USD. 1-year Forward $F_1 = 7.8200$ HKD per USD. $r_{\text{HKD}} = 5.0\%$, $r_{\text{USD}} = 3.0\%$. Identify and execute the covered interest arbitrage using a USD 10M borrowing notional. **Solution:**

- ① Calculate Theoretical CIP Forward Rate F^* :

$$F^* = 7.8000 \times \frac{1 + 0.050}{1 + 0.030} = 7.8000 \times \frac{1.05}{1.03} = 7.9515 \text{ HKD per USD}$$

- ② Actual Market Forward $F_{\text{market}} = 7.8200$ is below $F^* \implies$ USD is cheap forward relative to the interest differential.

- ③ Arbitrage Execution Strategy:

- ① Borrow USD 10,000,000 at 3%; owe USD 10,300,000 in one year.
 - ② Convert to HKD at spot: $\text{USD } 10\text{M} \times 7.8000 = \text{HKD } 78,000,000$.
 - ③ Invest HKD at 5%; receive $\text{HKD } 78\text{M} \times 1.05 = \text{HKD } 81,900,000$.
 - ④ Buy USD 10.3M forward at 7.8200, costing $\text{HKD } 10.3\text{M} \times 7.8200 = \text{HKD } 80,546,000$, and use the USD to repay the loan.
- ④ Net Profit = $\text{HKD } 81,900,000 - \text{HKD } 80,546,000 = +\text{HKD } 1,354,000$ before spreads, credit, and balance-sheet costs.

Problem: Investor borrows JPY 100M at $r_{\text{JPY}} = 0.1\%$ per annum, converts to AUD at $S_0 = 100.00$ JPY per AUD, invests at $r_{\text{AUD}} = 4.1\%$. 1 year later, spot rate moves to $S_1 = 92.00$ JPY per AUD (AUD depreciates 8%). Calculate net return in JPY. **Solution:**

- ① Convert initial JPY 100M to AUD: $\frac{100,000,000}{100.00} = \text{AUD } 1,000,000$.
- ② Invest in Australia for 1 year: $\text{AUD } 1,000,000 \times 1.041 = \text{AUD } 1,041,000$.
- ③ Convert back to JPY at $S_1 = 92.00$: $\text{AUD } 1,041,000 \times 92.00 = \text{JPY } 95,772,000$.
- ④ Repay JPY loan: $\text{JPY } 100,000,000 \times 1.001 = \text{JPY } 100,100,000$.
- ⑤ Net Payoff = $\text{JPY } 95,772,000 - \text{JPY } 100,100,000 = -\text{JPY } 4,328,000$, or -4.33% of the borrowed amount. An equity return would require the investor's equity contribution.
- ⑥ **Takeaway:** The $+4.0\%$ interest differential was more than offset by AUD depreciation, illustrating unhedged carry downside.

Problem: A HK firm expects EUR 1M receivable in 180 days. $S_0 = 8.5000$ HKD per EUR, 180-day $F = 8.5842$ HKD per EUR. Annualized simple rates (Actual/360): $r_{\text{HKD}} = 4.0\%$, $r_{\text{EUR}} = 2.0\%$. Put Option Strike $K = 8.5000$ with premium HKD 0.10 per EUR. Compare net HKD received under Forward Hedge vs Money Market Hedge vs Option Hedge (if $S_{180} = 8.4000$). **Solution:**

① **Forward Hedge:** Receives EUR 1M $\times 8.5842 =$ HKD 8,584,200.

② **Money Market Hedge:**

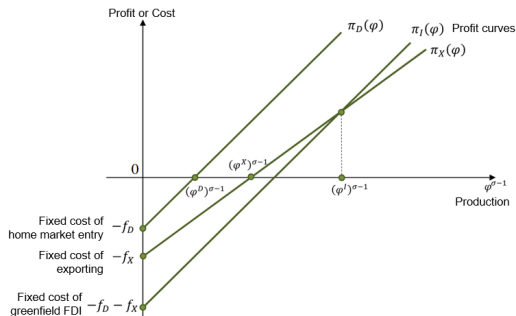
- Borrow EUR today: $\frac{\text{EUR 1M}}{1 + 0.02 \times \frac{180}{360}} =$ EUR 990,099.
- Convert to HKD at $S_0 = 8.5000 \implies$ HKD 8,415,842.
- Invest at HKD rate: $\text{HKD } 8,415,842 \times (1 + 0.04 \times \frac{180}{360}) =$ HKD 8,584,158.

③ **Option Hedge** (if $S_{180} = 8.4000 < K = 8.5000$): Exercise option at $K = 8.5000$.

- Gross HKD = EUR 1M $\times 8.5000 =$ HKD 8,500,000.
- Future value of option premium = EUR 1M $\times 0.10 \times (1 + 0.04 \times \frac{180}{360}) =$ HKD 102,000.
- Net HKD received = HKD 8,500,000 $-$ HKD 102,000 = HKD 8,398,000.

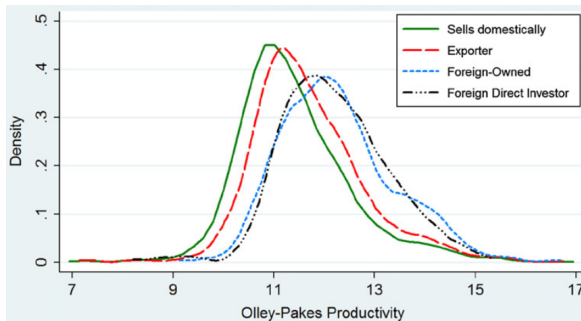
- Productive firms can cover the fixed costs of entering foreign markets. In benchmark heterogeneous-firm models, the most productive firms choose FDI, intermediate firms export, and others serve home markets or exit.
- This ordering requires assumptions about fixed costs, variable trade costs, and technology. Some real firms combine exporting, outsourcing, and affiliates.
- MNCs often concentrate research and coordination at headquarters while affiliates serve host markets. Distinguish all trade involving MNCs from transactions between related entities.
- Empirical distributions compare groups of firms. Productivity differences can reflect both selection into foreign markets and changes after entry.

Theoretical profit schedules: higher productivity can justify the fixed costs of exporting and FDI.



Source: Antras and Yeaple (2014).

Spain, 2007: productivity distributions for domestic firms, exporters, and MNCs. The distributions reflect selection into foreign activity as well as other differences between firms.

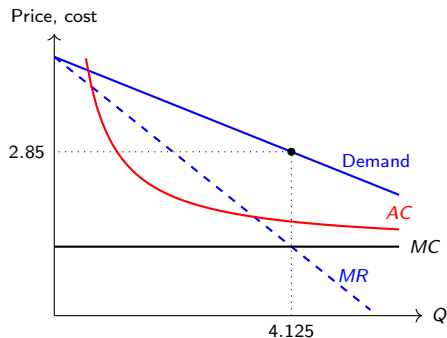


Source: Antras and Yeaple (2014).

Demand is $P = a - bQ$. Producing Q costs $cQ + F$.

- Selling more requires a lower price, so marginal revenue is $MR = a - 2bQ$.
- Set $MR = c$ to find $Q^* = (a - c)/(2b)$, then use demand to find the price.
- For $a = 4.5$, $b = 0.4$, $c = 1.2$, and $F = 1.8$:
 $Q^* = 4.125$, $P^* = 2.85$, and $AC^* = 1.636$.

Profit is $(P^* - c)Q^* - F = 5.006$.

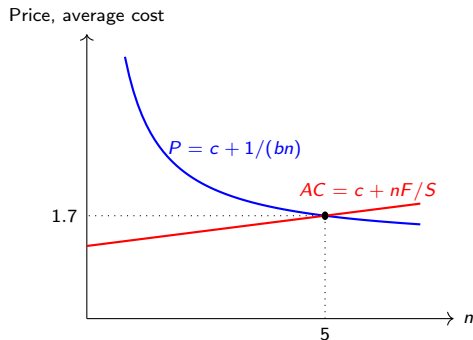


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Suppose n symmetric firms share a market of size S . Each pays fixed cost F .

- Average cost rises with the number of firms:
 $AC = c + nF/S$.
- Competition lowers the price: $P = c + 1/(bn)$.
- Free entry brings price down to average cost, giving $n^* = \sqrt{S/(bF)}$.

For $c = 1.2$, $b = 0.4$, and $F/S = 0.1$, the equilibrium is $n^* = 5$ and $P^* = 1.7$.

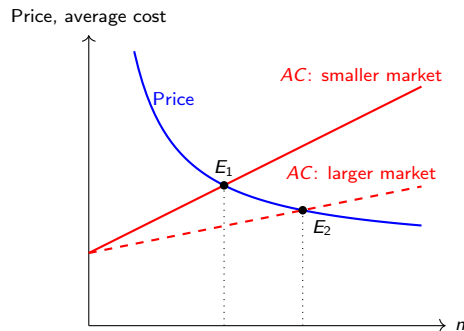


This stylized model assumes n is continuous. In reality, the number of firms is a natural number.

A larger market lets each firm spread its fixed cost over more sales.

- The average-cost curve shifts down. Entry rises, and competition lowers prices.
- Keep $c = 1.2$ and $b = 0.4$. If F/S falls from 0.5 to 0.2, n^* rises from 2.236 to 3.536.
- The equilibrium price falls from 2.318 to 1.907: about an 18% reduction in this example.

This is one motive for market integration. The EU, USMCA, and ASEAN differ greatly in how far they remove barriers.



- Market capitalization equals share price times shares outstanding on a specified date. Revenue and net income measure flows over a reporting period.
- For valuation, distinguish equity value from enterprise value. A common enterprise-value measure adds debt and subtracts excess cash, with adjustments for other claims.
- Comparisons among technology, energy, and manufacturing MNCs require common periods, currencies, and accounting definitions. Use each profit measure consistently: operating income, EBITDA, and net income deduct different expenses.
- Geographic revenue can be classified by customer location. Production location, incorporation, and taxable income require separate information.

The Census Bureau reports related-party goods trade for 2024:

	Related-party value, USD B	Share of category
Imports for consumption	1,610.8	49.5%
Total exports	727.2	35.3%
Combined goods trade	2,338.0	44.0%

- Related-party transactions accounted for 44.0% of US goods trade in 2024.
- The legal definition, denominator, and year matter. Goods crossing the border and local sales by foreign affiliates measure different activities.

Source: US Census Bureau, Related-Party Goods Trade, 2024.

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Amounts are USD billions, using each company's fiscal 2025 GAAP results.

Company	Global revenue	Net income	Fiscal year ended
NVIDIA	130.497	72.880	January 26, 2025
Microsoft	281.724	101.832	June 30, 2025

- Microsoft's revenue by customer location was USD 144.546B in the US and USD 137.178B elsewhere. These figures classify sales by customer location.
- Net income differs from operating income, EBITDA, and operating cash flow. Fiscal years also end on different dates.
- Calculating market capitalization requires share price and shares outstanding at a specified date.

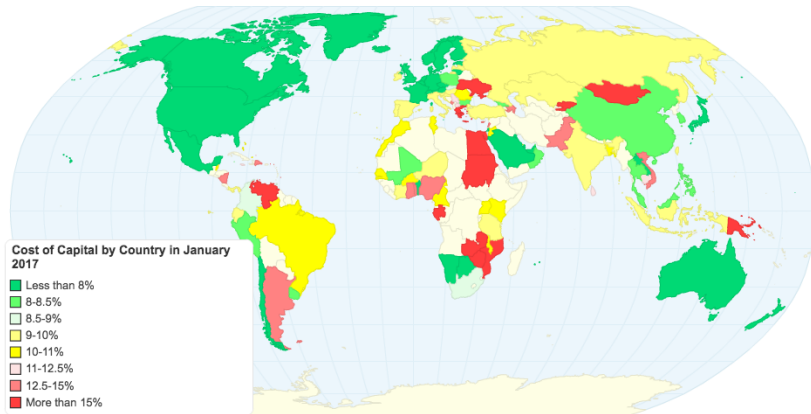
Sources: NVIDIA FY2025 results; Microsoft FY2025 annual report.

Stylized pre-2017 tax-rate exercise, with USD millions. Initial US profit is 400 and Irish profit is 50. Assume US tax 35%, Irish tax 12.5%, and no residual taxes.

Scenario	US tax	Irish tax	Total tax
Baseline	140.00	6.25	146.25
Shift 150 of royalties to Ireland	87.50	25.00	112.50
Also deduct Irish interest of 50	87.50	18.75	106.25

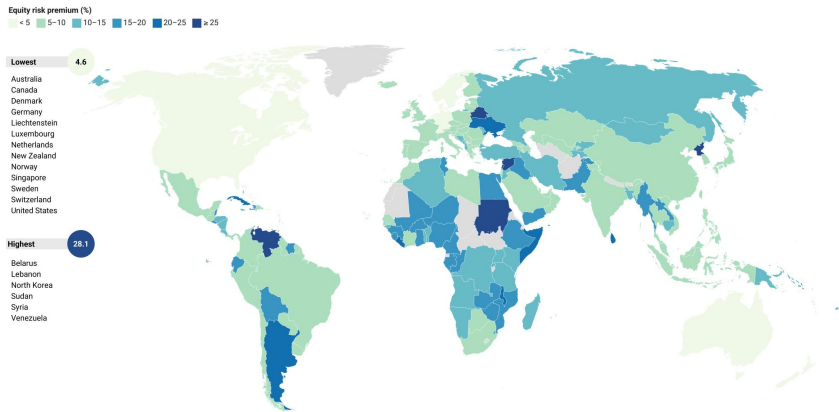
- Royalty shifting saves 33.75. The extra deduction saves at most 6.25 only if the recipient's interest income bears no tax. Combined saving is then 40.00.
- If the lender pays tax on the interest, subtract that tax from the saving. The baseline must retain the same real Irish profit of 50.
- This exercise uses historical tax assumptions. Arm's-length pricing, interest limitations, anti-hybrid rules, and residual taxes can eliminate the apparent benefit.

Source: OECD Transfer Pricing Guidelines (2022).



Historical exhibit. Data period, coverage, and units are as labeled.

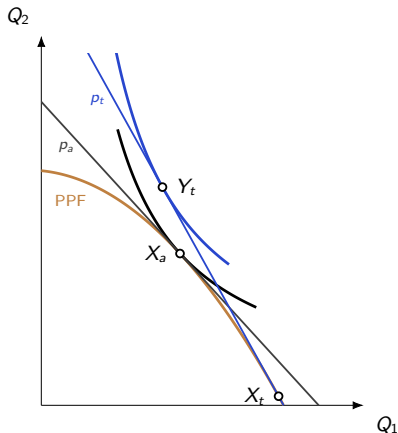
Source: Damodaran (2017).



Historical exhibit. Data period, coverage, and units are as labeled.

Source: Damodaran (2024).

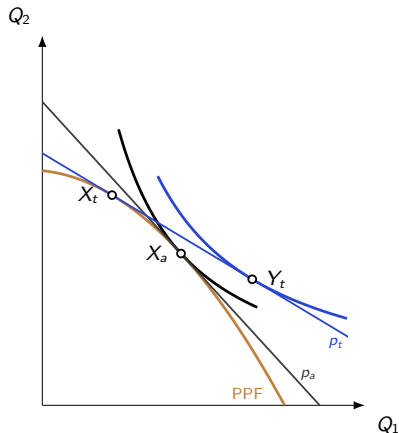
- The benchmark combines internationally traded goods with domestic factor markets. Labor, land, and capital are inputs, whose mobility is restricted in the model.
- Perfect competition, constant returns, and no distortions make the gains-from-trade argument transparent. The economy can afford its autarky consumption bundle at free-trade prices.
- Trade expands aggregate consumption possibilities when relative prices differ. Distributional losses, adjustment costs, and pre-existing distortions require separate analysis.
- Deardorff's general law describes an aggregate relationship across goods: countries tend to export goods with relatively low autarky prices.
- MNC models add fixed costs, differentiated products, firm productivity differences, and cross-border ownership to this benchmark.



- The brown **production possibilities frontier** (PPF) describes feasible domestic production.
- In autarky, production and consumption coincide at X_a . The black indifference curve and gray price line are tangent to the PPF.
- The world price ratio $p_t = p_1/p_2$ is higher than the autarky ratio p_a . The steeper blue price line makes good 1 relatively valuable. Home produces more good 1 at X_t , exports good 1, and imports good 2.
- Consumption at Y_t lies on a higher blue indifference curve. The difference between X_t and Y_t is the trade bundle.

Other comparative-advantage case

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- The brown **production possibilities frontier** (PPF) describes feasible domestic production.
- In autarky, production and consumption coincide at X_a . The black indifference curve and gray price line are tangent to the PPF.
- The world price ratio $p_t = p_1/p_2$ is lower than the autarky ratio p_a . The flatter blue price line makes good 2 relatively valuable. Home produces more good 2 at X_t , exports good 2, and imports good 1.
- Consumption at Y_t lies on a higher blue indifference curve. The difference between X_t and Y_t is the trade bundle.

[Other comparative-advantage case](#)[← Back to Main Slide](#)

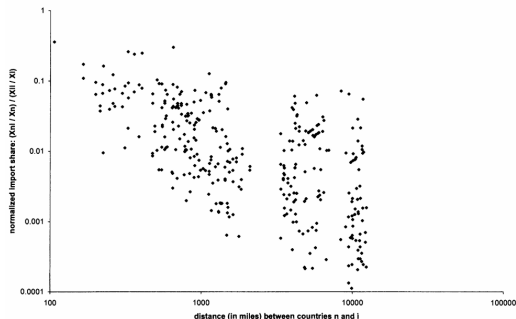


FIGURE 1.—Trade and geography.

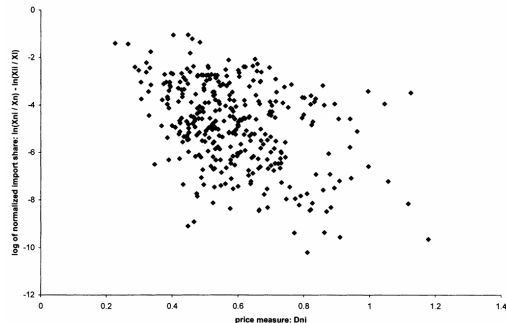


FIGURE 2.—Trade and prices.

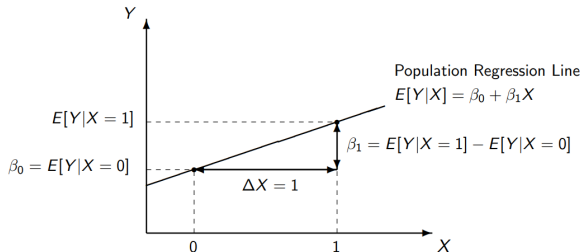
These exhibits accompany the Eaton–Kortum model. Technology and geographic trade costs jointly affect bilateral trade shares. The same market share can reflect different combinations of the two.

Source: Eaton and Kortum (2002).

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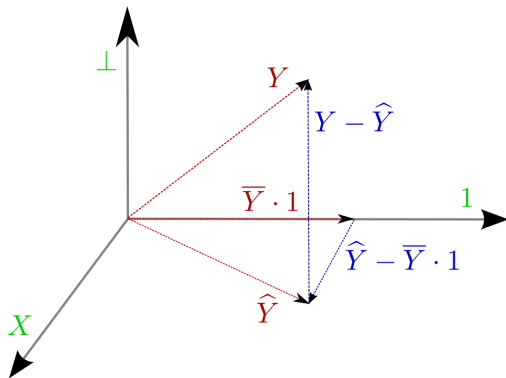
If X is binary, the population regression function is linear:

$$E[Y|X] = \beta_0 + \beta_1 X.$$



The conditional expectation $E[y | X]$ minimizes mean squared prediction error among all functions of X . The best linear predictor restricts the prediction to $X'\beta$. They coincide when the conditional expectation is linear in the included regressors.

$$\beta = E[XX']^{-1}E[XY], \quad \text{provided } E[XX'] \text{ is invertible.}$$



Source: Stanford Stats 191.

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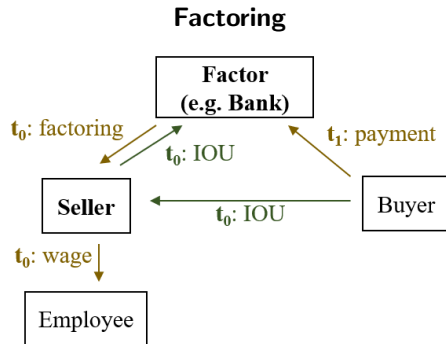
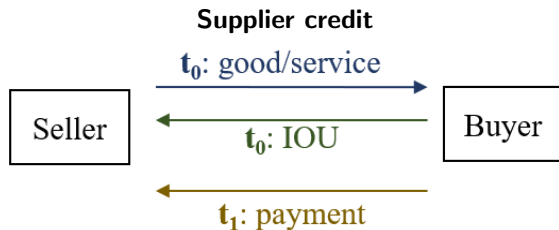
OLS chooses the fitted vector $\mathbf{X}\hat{\beta}$ closest to Y in squared Euclidean distance:

$$\hat{\beta} = \arg \min_b \|Y - \mathbf{X}b\|^2.$$

The residual is orthogonal to every included regressor:

$$\mathbf{X}'(Y - \mathbf{X}\hat{\beta}) = 0.$$

A causal interpretation requires assumptions about how the regressors relate to unobserved influences.



Supplier credit postpones the buyer's payment to the seller. Factoring introduces a financial intermediary that advances funds against the seller's receivable. The invoice remains tied to the underlying trade transaction.

Method of payment	Receipt of goods by buyer before payment	Settlement date	Risk rating for seller	Risk rating for buyer
Open account	✓	As per contract terms	Highest	Lowest
Usance collection	✓	At maturity as per collection terms	Medium to high	Low to medium
Sight collection	X	After presentation of documents	Lower	Low to medium
Documentary credit	X	After presentation of documents at sight or maturity as per contract terms	Low (against compliant documents)	High
Payment in advance	X	Prior to shipment	Lowest	Highest

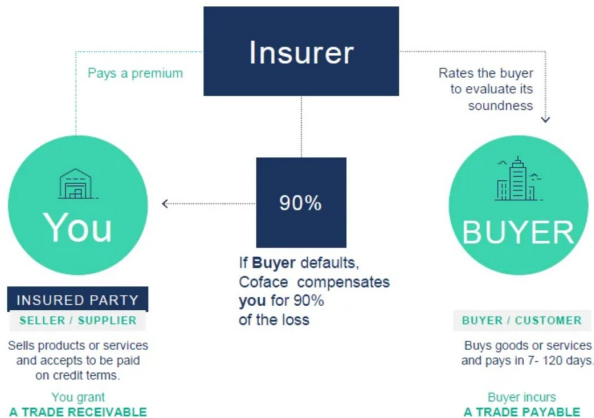
Source: www.tradefinance.training

Source: Deutsche Bank trade-finance reading.

RECOURSE FACTORING	NON-RECOURSE FACTORING
 Business responsible for unpaid invoices	 Factor responsible for unpaid invoices
 Credit risk held by the business	 Credit risk transferred to the factor
 Lower cost	 Higher cost
 Best fit for businesses with reliable customers	 Best fit for businesses seeking to reduce risk
 Disputes and other issues	 Disputes, deductions, and returns

The contract determines which losses the factor assumes. Non-recourse coverage may exclude disputes, fraud, ineligible invoices, or breaches of the seller's representations. Financing and risk transfer should be evaluated separately.

Source: Porter Capital illustration.



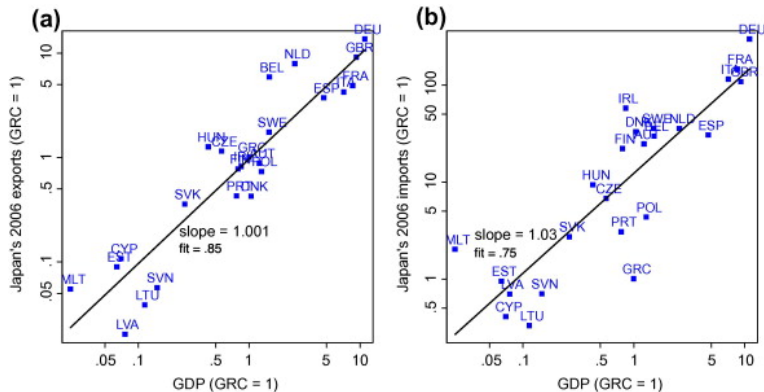
Insurance protects against specified non-payment risks, subject to exclusions and policy limits.

Source: Coface illustration.

In the simple labor-only Eaton–Kortum model,

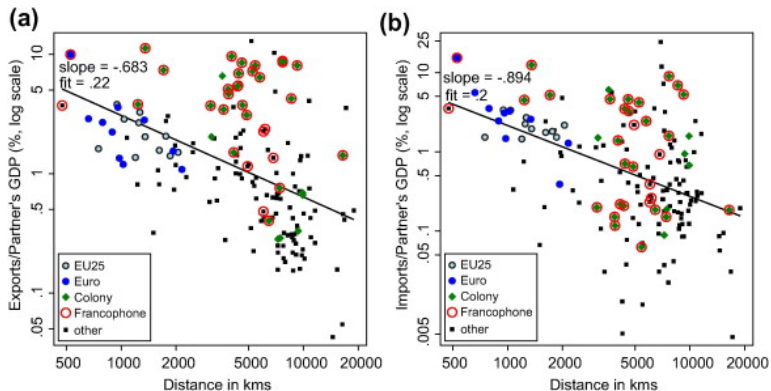
$$\pi_{ni} = \frac{T_i(w_i d_{ni})^{-\theta}}{\sum_j T_j(w_j d_{nj})^{-\theta}}, \quad X_{ni} = \pi_{ni} X_n.$$

- T_i captures technology, w_i unit labor cost, $d_{ni} \geq 1$ iceberg trade costs, and $\theta > 0$ the trade elasticity.
- A lower delivered cost increases an exporter's market share, holding other terms fixed. Competitors' costs also matter through the denominator.
- In this benchmark, real income is proportional to $T_n^{1/\theta} \pi_{nn}^{-1/\theta}$. Holding technology fixed, a lower domestic expenditure share raises real income.



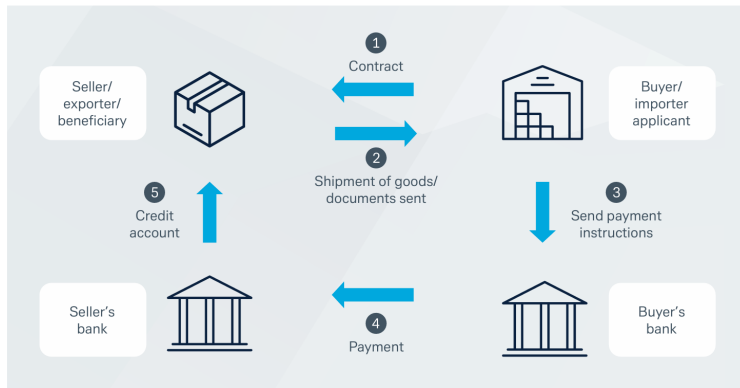
Historical exhibit from Head and Mayer (2014).

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Historical exhibit from Head and Mayer (2014).

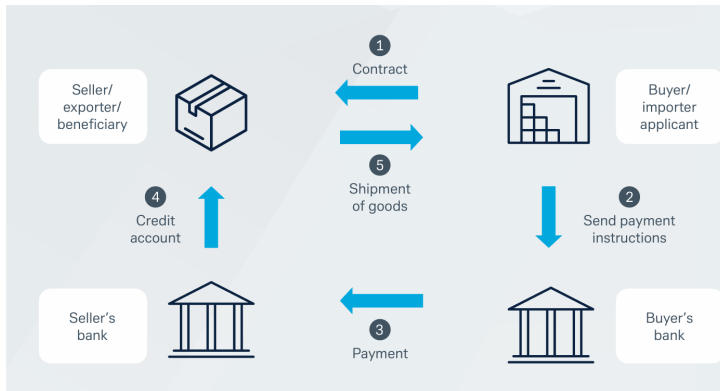
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Source: TradeLC Advisory

Seller ships first and extends credit. Non-payment can lead to contractual recovery action as well as loss of future business.

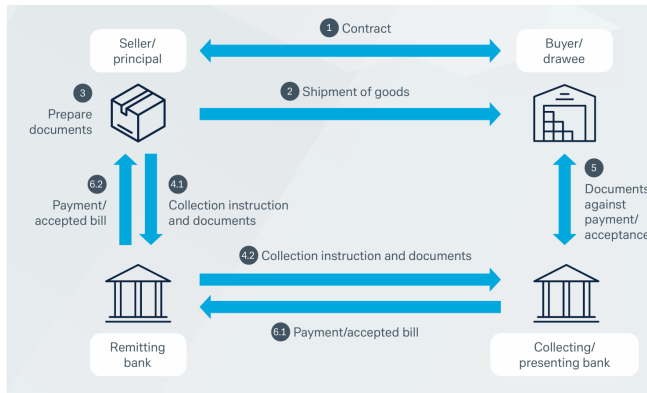
Diagram: Deutsche Bank.



Source: TradeLC Advisory

Buyer pays first and bears performance risk.

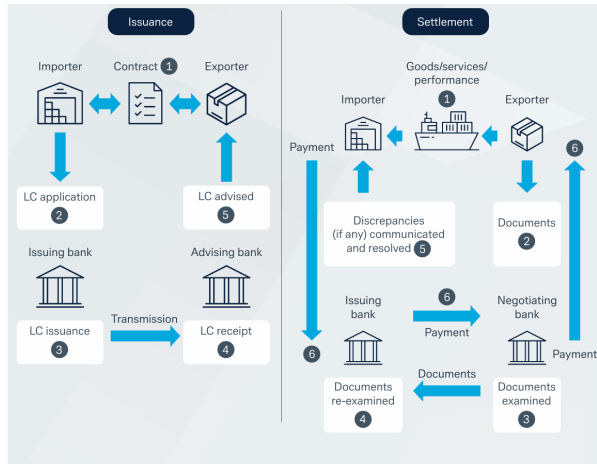
Diagram: Deutsche Bank.



Source: David Meynell

Banks handle documents and payment instructions. Neither D/P nor D/A creates a bank promise to pay.

Diagram: Deutsche Bank.

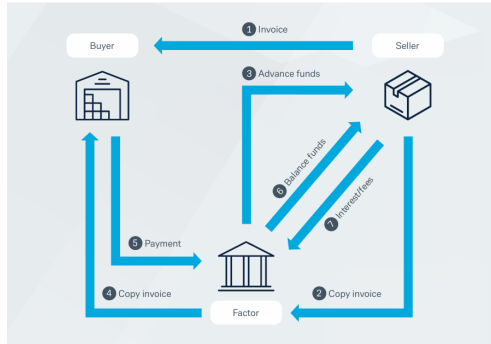


Source: TradeLC Advisory

Diagram: Deutsche Bank.

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Figure 14: Summary of factoring transactional flow

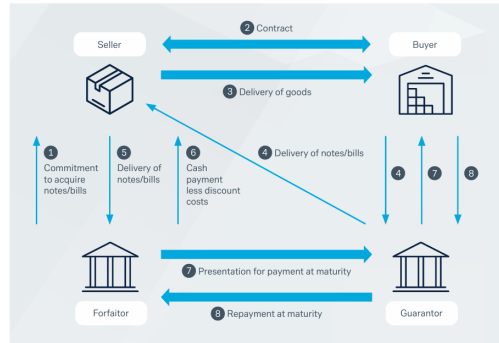


Source: TradeLC Advisory

Factoring often finances a stream of invoices. Forfeiting commonly purchases specified longer-term payment obligations.

Diagrams: Deutsche Bank.

Figure 13: Summary of forfeiting transactional flow



Source: www.tradefinance.training

- With recourse factoring, the exporter may have to repay the factor if the buyer defaults. With non-recourse factoring, the factor bears the defined credit risk within contractual limits.
- Credit insurance indemnifies covered losses after the policy conditions and waiting period are met. The exporter still finances the receivable unless a lender advances funds.
- Commercial default and specified political risks may be covered. Goods damage, quality disputes, and non-compliant documentation can fall outside cover.
- Compare the advance rate, discount fee, insurance premium, uncovered loss, and recovery timing. Risk transfer and accounting derecognition are separate questions.

Source: ITA, Export Credit Insurance.

Use the host-country perspective and describe gross assets and liabilities explicitly.

Transaction	Direct-investment entry	Counterpart / interpretation
Parent injects 100	Host FDI liability increases 100	Settlement changes another financial asset or liability
Affiliate earns 10 and retains 6	Reinvested earnings increase FDI liability by 6	Primary-income debit includes the attributable earnings
Affiliate pays dividend 4	Ordinary dividend is primary income	Settlement can occur through bank deposits
Parent buys existing shares for 100	FDI ownership changes	Purchase transfers ownership of existing capital

Source: IMF, BPM6, chapters 6, 8 and 11.

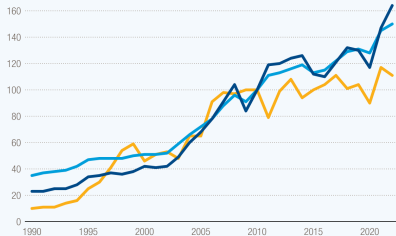
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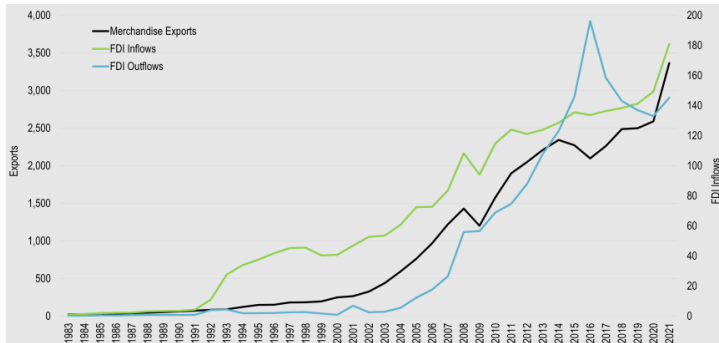
Foreign investment struggles to keep up with production and trade

Foreign direct investments (FDI), gross domestic product (GDP) and trade trends, indexed 2010 = 100

Underlying FDI trend GDP Trade



Source: UNCTAD, based on Eora26 and Asian Development Bank (ADB) input-output tables.
Note: Trade is global exports of goods and services. The underlying FDI trend is an UNCTAD indicator capturing the long-term dynamics of FDI by netting out fluctuations driven by one-off transactions and volatile financial flows (Vujanović et al., 2021).



Trade and FDI can move together because of market size, supply chains, and common policies.

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- A simple diminishing-returns model predicts high marginal products of capital where capital per worker is low. The Lucas paradox asks why observed international flows are smaller than that prediction.
- Human capital, infrastructure, institutions, and technology can complement physical capital.
- Political risk, weak contract enforcement, and information problems can separate a high physical return from the return foreign investors receive.
- For FDI, local financing and supplier capabilities affect both entry profitability and spillovers. A parent's internal funds solve only some of these problems.

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Exporting (X) and a foreign affiliate (I) sell the same quantity q at the same price p . Each pays a fixed cost F and a variable cost c per unit.

1. Write profits: $\pi_X = (p - c_X)q - F_X$, $\pi_I = (p - c_I)q - F_I$.

2. Subtract exporting: $\pi_I - \pi_X = (p - c_I)q - F_I - (p - c_X)q + F_X$
 $= (c_X - c_I)q - (F_I - F_X).$

3. Set the difference to zero: $(c_X - c_I)q^* = F_I - F_X.$

4. Solve for sales: $q^* = \frac{F_I - F_X}{c_X - c_I}.$

5. Interpret: when $F_I > F_X$ and $c_X > c_I$, choose FDI above q^* , exporting below q^* , and either at q^* . This threshold compares the two modes; both could still lose money.

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2. Subtract exporting: $\pi_I - \pi_X = (p - c_I)q - F_I - (p - c_X)q + F_X$
 $= (c_X - c_I)q - (F_I - F_X).$

3. Set the difference to zero: $(c_X - c_I)q^* = F_I - F_X.$

4. Solve for sales: $q^* = \frac{F_I - F_X}{c_X - c_I}.$

5. Interpret: when $F_I > F_X$ and $c_X > c_I$, choose FDI above q^* , exporting below q^* , and either at q^* . This threshold compares the two modes; both could still lose money.

- ① **Extra fixed cost:** $F_I - F_X = \text{USD } 8,000,000$ for the affiliate.
- ② **Saving per unit:** $c_X - c_I = \text{USD } 20$. The selling price is the same, so each affiliate sale contributes USD 20 more toward fixed cost.
- ③ **Set total savings equal to extra fixed cost:**

$$20q^* = 8,000,000.$$

- ④ **Divide by the per-unit saving:**

$$q^* = \frac{8,000,000 \text{ USD}}{20 \text{ USD/unit}} = 400,000 \text{ units.}$$

- ⑤ **Check the choice** using $\pi_I - \pi_X = 20q - 8,000,000$:

Sales	Affiliate profit minus export profit	Preferred mode
300,000	−USD 2M	Export
400,000	0	Indifferent
500,000	+USD 2M	FDI

Use the same perpetual asset cash flows: 150 or 50, asset value 1,000, and debt 200 paying 10 each year. Suppose levered equity trades at 850, so $V_L = 1,050 > V_U$.

Trade	Cash today	High payoff	Low payoff
Short levered equity	+850	-140	-40
Buy unlevered assets	-1,000	+150	+50
Borrow personally	+200	-10	-10
Total	+50	0	0

The investor receives 50 immediately and has no net future liability. This requires identical asset payoffs, borrowing terms, and unrestricted short sales.

Keep the same asset and debt cash flows, but suppose levered equity trades at 750. Then $V_L = 950 < V_U$.

Trade	Cash today	High payoff	Low payoff
Short unlevered assets	+1,000	-150	-50
Buy levered equity	-750	+140	+40
Buy the firm's debt	-200	+10	+10
Total	+50	0	0

Buying the firm's actual debt matches its payoff, including any default states.

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All amounts are millions. The exchange rate is 1.10 USD per 1 EUR and the USD discount rate is 10%.

Step	Amount
Revenue less cash costs	$30 - 18 = 12$ EUR
Less depreciation	$12 - 5 = 7$ EUR
Host tax at 25%	1.75 EUR
Operating cash after tax	$5.25 + 5 = 10.25$ EUR
Cash after 10% withholding	$10.25(0.90) = 9.225$ EUR
Convert to USD	$9.225(1.10) = 10.1475$ USD
Discount to date 0	$10.1475/1.10^1 = 9.2250$ USD

All amounts are millions. The exchange rate is 1.10 USD per 1 EUR and the USD discount rate is 10%.

Step	Amount
Revenue less cash costs	$32 - 19 = 13$ EUR
Less depreciation	$13 - 5 = 8$ EUR
Host tax at 25%	2.00 EUR
Operating cash after tax	$6.00 + 5 = 11.00$ EUR
Cash after 10% withholding	$11.00(0.90) = 9.900$ EUR
Convert to USD	$9.900(1.10) = 10.8900$ USD
Discount to date 0	$10.8900/1.10^2 = 9.0000$ USD

All amounts are millions. The exchange rate is 1.10 USD per 1 EUR and the USD discount rate is 10%.

Step	Amount
Revenue less cash costs	$34 - 20 = 14$ EUR
Less depreciation	$14 - 5 = 9$ EUR
Host tax at 25%	2.25 EUR
Operating cash after tax	$6.75 + 5 = 11.75$ EUR
Add salvage (equals tax book value)	12.00 EUR
Cash after 10% withholding	$23.75(0.90) = 21.375$ EUR
Convert to USD	$21.375(1.10) = 23.5125$ USD
Discount to date 0	$23.5125/1.10^3 = 17.6653$ USD

The exercise applies withholding to all terminal cash. If sale proceeds differ from tax book value, include tax on the gain or loss before remittance.

EUR millions in Year 1	Baseline	Additional parent markup of 4
Subsidiary revenue	30.00	30.00
Subsidiary cash costs	18.00	22.00
Depreciation	5.00	5.00
Taxable profit	7.00	3.00
Host tax at 25%	1.75	0.75
Cash before withholding	10.25	7.25
Dividend after 10% withholding	9.225	6.525
Additional parent income after 20% tax	0	3.200
Combined parent receipts	9.225	9.725

Consolidated receipts increase by EUR 0.5M.

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Hypothetical pre-2017 tax exercise, USD millions. An Irish affiliate pays a US parent for intellectual property. Assume 35% US tax, 12.5% Irish tax, and no residual home tax.

- Benchmark royalty: 200. Reduced royalty: 50. Paying 150 less raises Irish profit by 150 and lowers US profit by 150.
- US profit falls from 400 to 250. Irish profit rises from 50 to 200. Group pretax profit remains 450.
- Total tax falls from $400(0.35) + 50(0.125) = 146.25$ to $250(0.35) + 200(0.125) = 112.50$.
- The arithmetic saving is $150(0.35 - 0.125) = 33.75$.

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Continue the hypothetical historical royalty exercise. The Irish unit borrows USD 1,000M at 5%, creating annual interest of USD 50M.

	No additional interest	With interest
Irish profit before interest	200.00	200.00
Irish taxable profit	200.00	150.00
Irish tax at 12.5%	25.00	18.75

- The Irish deduction saves 6.25. If the lender pays tax t_L on the 50 received, the net group saving is $6.25 - 50t_L$.
- Historical structures routed interest to low-tax or differently treated entities.
- Reverse the loan direction and the deduction changes too. A US borrower paying an Irish lender has different tax arithmetic from an Irish borrower paying a US lender.

- The historical Renault–Nissan alliance combined cross-shareholdings, shared management, and state influence through the French government's stake in Renault.
- Cash-flow ownership and voting control can differ from each partner's economic contribution. A partner can perceive its influence as too small relative to its sales or technology contribution.
- Joint decisions about leadership, investment, and electric vehicles can therefore create conflicts among partners and their shareholders.
- **Discussion:** which decisions belong to each company's board, and which require agreement across the alliance? What protections would minority investors want?

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- Enron's use of complex special-purpose entities and related-party arrangements obscured debt, risks, and the quality of reported earnings.
- The governance problem was misleading reporting and conflicts of interest. Evaluate these issues within each entity.
- Creditors and shareholders need to identify guarantees, related-party exposures, and the economic substance of transactions across the group.
- **Discussion:** which reported profits would you reconcile to cash, and what obligations might sit outside the apparent operating balance sheet?

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- The Samsung SDS example discussed in Bae, Kang, and Kim (2002) concerns securities issued to members of a controlling family on terms criticized as favorable to the recipients.
- The historical case contrasts a subscription price of KRW 7,150 with an indicated outside-market price of KRW 55,000 for the relevant share entitlement.
- The economic question is whether the transaction transfers value away from outside shareholders. Valuation date, security terms, and allocation rights must be comparable.
- Pyramids and cross-holdings can separate control from cash-flow ownership. Independent review and minority approval can constrain abusive transactions.

Source: Bae, Kang, and Kim (2002), historical case.

- The Vale example asks how a foreign investor can monitor a geographically dispersed business governed by Brazilian institutions.
- Physical distance, language, operational complexity, and local enforcement can raise the cost of evaluating management and coordinating shareholders.
- Assess Vale using its reporting framework, filings, and Brazilian enforcement institutions.
- **Discussion:** how would local expertise, site-level disclosure, independent directors, and external assurance change your willingness to provide capital?

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- A two-tier system separates management from a supervisory board. Representation by major shareholders, employees, or public bodies can create competing objectives.
- Volkswagen's emissions scandal illustrates that formal oversight structures can fail to prevent misconduct. The case raises questions about board information, incentives, and enforcement.
- In January 2017, Volkswagen agreed to plead guilty and pay USD 4.3B in US criminal and civil penalties. Individual executives and employees also faced charges.
- **Valuation link:** misconduct can create fines, remediation costs, lost sales, and financing constraints across multiple jurisdictions.

Source: US Department of Justice, January 11, 2017.

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- An ADR allows investors to hold an interest in a foreign issuer's shares through a depositary arrangement. Exchange-listed programs bring disclosure and regulatory obligations.
- Petrobras's access to US investors exposed its disclosures to US securities enforcement. In 2018, the SEC announced a settlement over misleading investors.
- This illustrates both the potential value and the limits of regulatory bonding. Additional oversight can sanction wrongdoing without preventing every abuse.
- **Discussion:** how could changes in disclosure and enforcement affect investors' required return after a cross-listing?

Source: SEC, Petrobras settlement, September 27, 2018.

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- La Porta, Lopez-de-Silanes, Shleifer, and Vishny compared investor protections and ownership concentration across countries with different legal origins.
- Their historical findings associate common-law systems with stronger measured shareholder protection than some civil-law groups, including the French-origin group.
- The mechanism is financing: outside investors may supply less capital or demand control when they expect expropriation and weak remedies.
- The result is an empirical association for particular samples and measures. Reforms, enforcement, ownership, and issuer-specific arrangements matter for a current investment decision.

Source: La Porta et al., Law and Finance.

- In 2008, coordinated US and German actions resulted in approximately USD 1.6B of penalties against Siemens for bribery-related misconduct.
- Cross-border business creates exposure to more than one enforcement system. A transaction may need to satisfy rules in several jurisdictions.
- Boards and managers need controls over agents, payments, approvals, and financial reporting. External audits and credible sanctions complement internal monitoring.
- **Valuation link:** expected penalties and disruption can reduce cash flows. Assess any financing-cost effect using separate evidence.

Source: US Department of Justice, December 15, 2008.

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Statistic in the Antràs–Yeaple reading	Historical observation
Top 1% of US exporters' share of US exports (2000)	81%
Foreign-affiliate share of large US firms' foreign sales (2009)	About 75%
US exports and imports involving multinational firms	Roughly 90%
US manufacturing affiliates' sales by destination (2009)	55% host, 34% third country, 11% US
Cross-border M&A as a share of FDI flows (2007)	68% developed, 18% developing

Trade involving an MNC includes sales to unrelated firms. These historical statistics describe the samples and periods reported.

Source: Antràs and Yeaple (2014), pp. 56, 65–66.

Percentage of host-country manufacturing accounted for by foreign affiliates.

	Finland	France	Ireland	Netherlands	Poland	Sweden
Enterprises	1.6	2.0	13.4	3.4	16.0	2.8
Employment	17.2	26.2	48.0	25.1	28.1	32.4
Sales	16.2	31.8	81.1	41.1	45.2	39.9
R&D	13.1	27.4	77.3	35.8	20.9	52.0
Exports	17.5	39.5	92.3	60.0	69.1	45.8

Affiliates' shares of employment and sales greatly exceed their shares of firms. The exact shares differ across countries, so a single "under 5%, over 30%" rule is misleading.

Source: OECD (2007), reproduced in Antràs and Yeaple (2014), Table 2.1. Historical manufacturing data.

These multinational firms span different businesses and asset requirements.

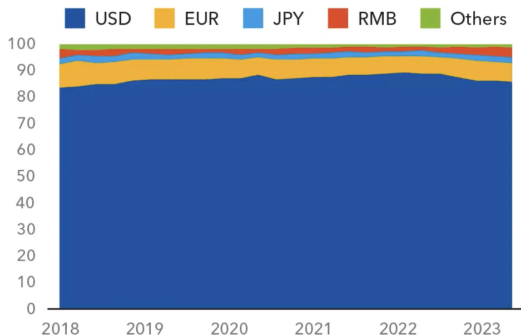
Examples	Valuation questions
NVIDIA, Broadcom, TSMC	IP ownership, fabrication, customers, and geographic production
Microsoft, Apple, Alphabet, Meta	Intangibles, platforms, and the location of sales versus taxable income
Amazon, Tesla	Logistics or production capacity, working capital, and margins
Saudi Aramco	Commodity prices, physical reserves, and capital expenditure
Berkshire Hathaway	Different operating businesses, investments, and financial claims

A descriptive log-linear specification is

$$\log X_{ni} = \alpha_n + \alpha_i + \zeta \log D_{ni} + \varepsilon_{ni}.$$

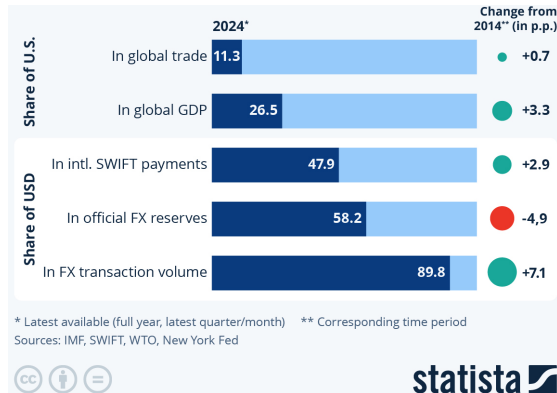
- Importer and exporter fixed effects absorb country-specific factors. A negative ζ associates greater distance with lower bilateral trade.
- If $\zeta = -0.9$, a 1% increase in distance is associated with about 0.9% less trade. Doubling distance implies a factor of $2^{-0.9} \approx 0.536$.
- OLS is a best linear prediction under its projection interpretation. A causal interpretation requires additional identification assumptions.
- Zero trade flows, heteroskedasticity, and omitted bilateral costs complicate log OLS. A gravity relationship can arise from several different trade models.

Source: Head and Mayer (2014).



Sources: COFER, SWIFT and staff calculations.
 Note: Only includes trade finance covered by SWIFT (Code MT700 (Letters of credit)).
 Data as of 2023Q3 (FX reserves), 2023Q4 (Trade Finance). Data is exchange-rate adjusted.

IMF



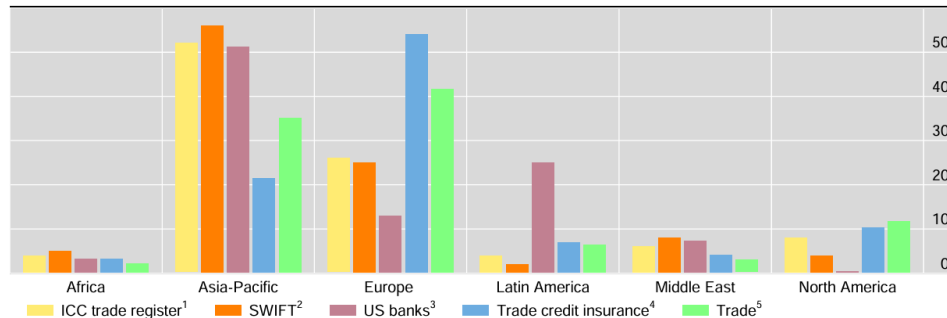
statista

A third currency (USD) can dominate invoicing and financing even when neither trading partner uses it domestically. The invoice currency affects working-capital and FX exposure.

IMF and Statista. Dates and measures differ across panels.

As a share of total, in per cent

Graph 2



¹ Average from 2008 to 2011. ² Based on average value of sent and received SWIFT MT700 messages in 2011. ³ The US data capture only lending vis-à-vis non-residents resulting in a low share of US banks' exposure to North America. Average from 2008 to September 2012. ⁴ Short term credit insurance from the Berne Union. Average for Q4 2011 to Q1 2013. ⁵ Merchandise trade (average of imports and exports) from Q1 2008 to Q4 2012.

Sources: Berne Union; ICC; IMF; SWIFT; national data; Group calculations.

Illustrative classroom example. All figures and assumptions are hypothetical.

An exporter compares markets A and B. Hold its own GDP and all other trade frictions fixed, and assume $\gamma = 1$.

Destination	GDP index	Distance index	Predicted exports
A	100	1	USD 20M
B	200	2	USD 20M
B after lower effective trade costs	200	1.5	USD 26.67M

Initially, $X_B/X_A = (200/100)(1/2) = 1$. After the change, $X_B/X_A = 2/1.5 = 1.333$.

Interpretation

A larger market can be offset by higher trade costs. This model calculation holds all other determinants fixed.

Illustrative classroom example. All figures and assumptions are hypothetical.

An exporter pays USD 1M on day 0 and collects USD 1.08M on day 90. Borrowing costs 8% simple annual interest on an Actual/360 basis.

Collection date	Interest cost	Cash left after repayment
Day 90	$1,000,000(0.08)(90/360) = 20,000$	USD 60,000
Day 120	$1,000,000(0.08)(120/360) = 26,667$	USD 53,333

Interpretation

A 30-day delay consumes USD 6,667 of margin even if the buyer eventually pays in full. Funding the extra month requires cash or additional borrowing, even with credit insurance.

Illustrative classroom example. All figures and assumptions are hypothetical.

A coffee exporter sells a USD 100,000 shipment. It must pay suppliers before shipment.

- **Cash in advance:** the buyer pays before shipment and bears the risk that the exporter fails to deliver.
- **Open account:** the exporter ships now and receives payment in 60 days, bearing buyer non-payment risk and financing the delay.
- **After ten successful orders:** open account may become acceptable if the exporter can fund the receivable and assess the buyer's credit.

Interpretation

Settlement terms allocate risk and financing between the parties. A longer relationship can change the choice without eliminating country or delivery risk.

Illustrative classroom example. All figures and assumptions are hypothetical.

An exporter ships USD 500,000 of machinery. The buyer becomes short of cash after shipment.

- **Documents against payment:** the collecting bank releases the documents only against payment. If the buyer refuses, the exporter retains the payment risk.
- **Letter of credit:** if the presentation complies, the issuing bank must honor its undertaking. The exporter still faces bank and country risk.
- **Document discrepancy:** a late bill of lading can prevent payment under the credit unless the discrepancy is resolved or waived.

Interpretation

A bank's payment undertaking is the key difference. The bank checks documents. Product quality requires separate inspection.

Illustrative classroom example. All figures and assumptions are hypothetical.

An exporter holds a USD 100,000 invoice due in 90 days. Ignore additional fees and claim delays.

- **Factoring:** a factor buys the full invoice for USD 97,000 today. The exporter gives up USD 3,000 to obtain cash immediately. **Recourse:** if the buyer defaults, the exporter must reimburse the factor under the assumed contract. Without recourse for covered insolvency, the factor bears that loss.
- **Insurance instead:** a USD 1,000 premium buys 90% cover. A covered total default produces a USD 90,000 indemnity, leaving USD 10,000 uninsured plus the premium.
- **Forfaiting:** for a five-year equipment sale, the exporter could instead sell eligible guaranteed payment claims without recourse.

Interpretation

Financing accelerates cash receipt. Insurance transfers specified losses. Contract exclusions and recourse determine who ultimately bears the risk.

Illustrative classroom example. All figures and assumptions are hypothetical.

A Hong Kong manufacturer sells HKD 30M of goods to an independent Malaysian distributor each year.

- **Before the acquisition:** all owned production and management remain in Hong Kong. The firm exports through an independent distributor.
- **After the acquisition:** the manufacturer buys 80% of the distributor and controls its operations. The group now operates in both economies.
- **New corporate-finance decisions:** the parent chooses equity or intercompany debt, supervises local managers, and evaluates cash remitted in another currency.

Interpretation

Acquiring control of the foreign distributor turns the exporter into an MNC.

Illustrative classroom example. All figures and assumptions are hypothetical.

In a hypothetical economy, total goods exports are USD 100B and total business sales are USD 400B.

Measure	Numerator	Denominator	Share
MNC share of exports	MNC exports: 60	All exports: 100	60%
Related-party share of exports	Intragroup exports: 25	All exports: 100	25%
Foreign-affiliate share of sales	Affiliate sales: 80	All sales: 400	20%

Interpretation

The shares answer different questions. Affiliate sales may include local sales, while related-party exports are shipments across a border. Compare each share using its own denominator.

Illustrative classroom example. All figures and assumptions are hypothetical.

Consider ordinary voting shares with no special control agreements. All investors and target firms are in different economies.

Transaction	Classification	New capacity from the transaction?
Buy 5% of voting shares	Portfolio equity	No
Buy 15% of voting shares	FDI relationship	No
Acquire 100% of an existing factory	FDI acquisition	No
Fund construction of a new owned factory	Greenfield FDI	Yes

Interpretation

The 10% voting threshold identifies a direct-investment relationship. Majority control and the creation of new productive capacity are separate questions.

Illustrative classroom example. All figures and assumptions are hypothetical.

An existing foreign-owned affiliate builds a USD 100M factory this year with financing:

Funding source	USD M	Inward FDI flow
New parent equity	40	40
Retained earnings attributable to parent	20	20
Loan from a local bank	40	0
Total	100	60

Illustrative classroom example. All figures and assumptions are hypothetical.

A medical-device firm expects 100,000 foreign sales at USD 100 each. Compare one-year profit before tax.

Mode	Unit cost	Annual fixed cost	Profit
Export	USD 70	USD 0.2M	USD 2.8M
Outsource abroad	USD 50	USD 0.5M	USD 4.5M
Own foreign factory	USD 40	USD 2.0M	USD 4.0M

If outsourcing also causes an expected USD 1M loss from defects or intellectual property (IP) leakage, its adjusted profit falls to USD 3.5M.

Interpretation

Compare total profit after variable costs, fixed costs, and expected losses from defects or IP leakage.

Illustrative classroom example. All figures and assumptions are hypothetical.

A clothing brand compares four ways to produce the same garment.

		Independent supplier	Owned operation
At home	Contract with a local workshop	Use its existing factory	
Abroad	Contract with a foreign workshop	Build a foreign affiliate	

- For standardized T-shirts, a contract may specify quality sufficiently well for offshore outsourcing.
- For a proprietary fabric, weak protection of process knowledge may justify ownership, even in the same foreign location.

Interpretation

Foreign production can use independent suppliers or owned affiliates.

Illustrative classroom example. All figures and assumptions are hypothetical.

A consumer-products group considers three foreign projects.

- **Cost:** a component plant saves USD 4 per unit on 2M units, but adds USD 3M of annual logistics costs. Net annual savings are USD 5M before setup costs.
- **Capability:** buying a distributor provides its brand and retail network. These distribution assets complement the group's production capacity.
- **Customers:** a local kitchen adapts recipes and delivers fresh food. Local preparation supports freshness and adaptation to customer preferences.

Diversification reduces risk when country cash flows are less than perfectly correlated.

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Illustrative classroom example. All figures and assumptions are hypothetical.

A differentiated producer has constant marginal cost USD 20 and annual fixed cost USD 600. At its chosen price of USD 30, it sells 100 units.

Situation	Sales	Contribution	Profit
Initial market	100	$(30 - 20)100 = 1,000$	USD 400
After competing varieties enter	60	$(30 - 20)60 = 600$	USD 0

For this calculation, hold the firm's price fixed while entry reduces its demand. A complete equilibrium can also change price.

Interpretation

A positive markup can occur with zero economic profit because the markup pays for fixed costs. Entry reduces incumbents' profits and expands consumer choice.

Illustrative classroom example. All figures and assumptions are hypothetical.

Three firms have the following annual contribution before foreign setup costs (USD M). Exporting costs 2 to enter, and FDI costs 6.

Firm productivity	Export contribution	FDI contribution	Best foreign choice
Low	1	3	Neither
Medium	4	7	Export: profit 2
High	8	15	FDI: profit 9

The medium firm earns $4 - 2 = 2$ from exports versus $7 - 6 = 1$ from FDI. The high firm earns 6 from exports versus 9 from FDI.

Interpretation

In this example, pre-existing productivity differences explain which firms choose FDI.

Illustrative classroom example. All figures and assumptions are hypothetical.

A host-country bank can supply USD 100M of loans. Local firms initially borrow the full amount.

Scenario	Affiliate local loan	Credit left for local firms
Affiliate borrows locally	USD 40M	USD 60M
Parent supplies affiliate funding	USD 0M	USD 100M
Local lending capacity rises to USD 150M	USD 40M	USD 110M

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Illustrative classroom example. All figures and assumptions are hypothetical.

Assume a sector caps foreign ownership at 50%.

- A foreign firm initially needs a joint-venture partner. Both parties contribute capital, and the agreement allocates decisions over technology, hiring, and dividends.
- If the cap is removed, the firm may choose full ownership and retain all residual cash flow, while also supplying all equity and managing the operation itself.
- The historical Tesla exhibit in the main deck illustrates the possibility of wholly foreign-owned entry after a sector liberalization.

Interpretation

A change in permitted ownership changes the feasible entry modes. Compare the NPVs of the available entry modes.

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Illustrative classroom example. All figures and assumptions are hypothetical.

A Brazilian affiliate receives BRL 10M next year. Compare two loans with the same BRL 5M proceeds today at 5 BRL per USD.

Loan	Promised repayment	At 5 BRL per USD	At 6 BRL per USD
BRL 5M at 10%	BRL 5.5M	BRL 5.5M	BRL 5.5M
USD 1M at 5%	USD 1.05M	BRL 5.25M	BRL 6.30M

Interpretation

Currency depreciation can outweigh the dollar loan's lower interest rate. BRL borrowing matches operating receipts, while a dollar loan introduces exchange-rate exposure. Compare hedged costs as well.

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Illustrative classroom example. All figures and assumptions are hypothetical.

A firm has equity with market value 80 and book value 20, and debt valued at 20 under either measure. Costs are $k_E = 12\%$, $k_D = 5\%$, and $\tau = 25\%$.

$$\text{WACC}_{\text{market}} = 0.8(12\%) + 0.2(5\%)(1 - 0.25) = 10.35\%.$$

$$\text{WACC}_{\text{book}} = 0.5(12\%) + 0.5(5\%)(1 - 0.25) = 7.875\%.$$

For a comparable-risk project costing 100 and paying 110 in one year, the NPVs are $110/1.1035 - 100 = -0.32$ and $110/1.07875 - 100 = +1.97$.

Interpretation

Book weights can reverse the accept/reject decision. Current market values measure investors' opportunity costs. The debt cost receives the tax adjustment once.

Illustrative classroom example. All figures and assumptions are hypothetical.

Use the main slide's perpetual assets: cash flow 150 or 50, each with probability one half, and unlevered value 1,000. Increase safe debt to 400 at 5%.

Claim	High state	Low state	Value
Debt	20	20	400
Equity	130	30	600
Total	150	50	1,000

$$k_E = 80/600 = 13.33\%, \quad \text{WACC} = 0.6(13.33\%) + 0.4(5\%) = 10\%.$$

Interpretation

Equity becomes riskier and demands a higher return. Under the frictionless assumptions, WACC and total firm value remain unchanged.

Illustrative classroom example. All figures and assumptions are hypothetical.

An unlevered business is worth 700. Assume permanent debt and fully usable tax shields valued at $0.20D$. All values use the same currency unit.

Debt	PV of tax shields	PV of distress costs	Firm value
0	0	0	700
200	40	20	720
400	80	45	735
600	120	90	730

Raising debt from 400 to 600 adds 40 in tax shields but 45 in expected distress costs.

Interpretation

Debt of 400 maximizes value among these choices. A subsidiary unable to use its deductions has a smaller tax benefit and may prefer less debt.

Illustrative classroom example. All figures and assumptions are hypothetical.

An affiliate needs USD 10M for a positive-NPV project. It has USD 4M of internal cash and can raise USD 6M of relatively safe debt.

- **Initial financing:** use 4 of internal cash and borrow 6, avoiding an equity sale when outside investors know less about the project.
- **If the project costs 13:** the extra 3 exceeds the assumed cash and debt capacity. Equity or a smaller project becomes necessary.
- **After a profitable year:** additional retained cash can finance the next project without restoring any particular debt ratio.

Interpretation

The observed debt ratio reflects funding needs and the history of retained earnings. This mechanism differs from maintaining a fixed leverage target.

Illustrative classroom example. All figures and assumptions are hypothetical.

A multinational issues USD 20M of debt after announcing an expansion.

Explanation	Additional evidence to look for
Trade-off theory	Debt moves leverage toward a stated target
Pecking order	Investment exceeds available internal cash
Currency matching	Debt service matches the affiliate's receipts

- If unexpectedly high retained earnings reduce next year's borrowing, that supports a financing-deficit explanation.
- If management instead issues debt to restore a target ratio, that supports a trade-off explanation.

Interpretation

The same financing action can fit several theories. The reason for the action and its response to new information help distinguish them.

Illustrative classroom example. All figures and assumptions are hypothetical.

An affiliate needs USD 10M now and expects USD 12M in one year. Its project required return is 10%. External lenders supply only USD 6M.

$$\text{NPV} = 12/1.10 - 10 = \text{USD } 0.91\text{M}.$$

- The parent can provide the missing USD 4M if transfers are permitted and this use exceeds the value of its alternatives.
- Intercompany funding reallocates group cash. If a guarantee replaces the transfer, the parent still bears the contingent liability.

Interpretation

Parent funding allows the affiliate to undertake a positive-NPV project. Evaluate this use of funds against alternative uses and transfer costs.

Illustrative classroom example. All figures and assumptions are hypothetical.

A project costs USD 80M and produces an expected USD 10M perpetuity. Assume segmentation alone raises its required return from 10% to 13%.

Market access	Required return	Project value	NPV
Segmented	13%	USD 76.92M	USD -3.08M
Integrated	10%	USD 100.00M	USD +20.00M

If gaining market access costs USD 3M today, the second NPV becomes USD 17M.

Interpretation

Lower financing barriers can make a project viable. Use the project-specific required return in each financing environment.

Illustrative classroom example. All figures and assumptions are hypothetical.

A project costs EUR 100 now and pays EUR 110 in one year. Let $S_0 = 1.20$ USD per 1 EUR, $k_f = 10\%$, and $k_h = 5\%$. Assume consistent risk-adjusted conversion.

$$\hat{S}_1 = 1.20 \frac{1.05}{1.10} = 1.14545.$$

$$\text{NPV}_{\text{EUR}} = -100 + \frac{110}{1.10} = 0.$$

$$\text{NPV}_{\text{USD}} = -120 + \frac{110(1.14545)}{1.05} = 0.$$

Interpretation

The two valuations agree under the specified pricing relation. Holding FX at 1.20 while mixing these discount rates would instead give a spurious USD 5.71 NPV.

Illustrative classroom example. All figures and assumptions are hypothetical.

A wholly owned affiliate generates EUR 10M of after-tax free cash flow in Year 1. It retains EUR 4M under a remittance limit and distributes EUR 6M.

Step	Amount
Dividend before withholding	EUR 6.00M
Withholding at 10%, with no usable credit	EUR 0.60M
Cash reaching parent	EUR 5.40M
Convert at 1.10 USD per EUR	USD 5.94M

The EUR 4M retained belongs in a separate forecast of local investment returns, later distributions, or terminal value.

Interpretation

This year's parent receipt differs from subsidiary free cash flow. Giving the retained funds zero value would omit an asset still owned by the group.

Illustrative classroom example. All figures and assumptions are hypothetical.

At the end of Year 1, EUR 5M is available for distribution. Assume constant FX, a 10% required return, no extra tax, and a certain release after two years.

Use of funds	Value at the end of Year 1
Remit immediately	EUR 5.000M
Retain at 2%, release in Year 3	$5(1.02)^2 / (1.10)^2 = \text{EUR } 4.299\text{M}$
Retain at a comparable-risk 10% return	$5(1.10)^2 / (1.10)^2 = \text{EUR } 5.000\text{M}$

Interpretation

The value loss is EUR 0.701M in the 2% case. A restriction has a smaller cost when local funds earn an appropriate risk-adjusted return.

Illustrative classroom example. All figures and assumptions are hypothetical.

A component division has spare capacity and marginal cost USD 60. An assembly division can finish the product for USD 20 and sell it outside the group for USD 100.

Internal transfer price	Assembly division margin	Group margin if produced
USD 70	$100 - 70 - 20 = 10$	$100 - 60 - 20 = 20$
USD 90	$100 - 90 - 20 = -10$	$100 - 60 - 20 = 20$

If the assembly manager can refuse a loss-making order, a USD 90 internal price can prevent production.

Interpretation

The group loses USD 20 of contribution despite a profitable outside sale.

Illustrative classroom example. All figures and assumptions are hypothetical.

Use hypothetical tax rates: parent 20%, subsidiary 30%. This time the parent produces for EUR 80 and the subsidiary resells outside the group for EUR 200.

Transfer price	Parent profit	Subsidiary profit	Combined tax
EUR 100	20	100	$0.20(20) + 0.30(100) = 34$
EUR 140	60	60	$0.20(60) + 0.30(60) = 30$

Interpretation

Raising the transfer price now shifts profit to the lower-tax parent and reduces tax by EUR 4. The direction reverses the main slide's result.

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Illustrative classroom example. All figures and assumptions are hypothetical.

A project has revenue 12, cash operating costs 6, depreciation 2, tax rate 20%, new CAPEX 1, and an increase in working capital of 0.5. All amounts are EUR M.

Step	Calculation	EUR M
EBIT	$12 - 6 - 2$	4.0
Operating profit after tax	$4(1 - 0.20)$	3.2
Operating cash flow	$3.2 + 2$	5.2
Free cash flow	$5.2 - 1 - 0.5$	3.7

Interpretation

Depreciation reduces taxes by EUR 0.4M. Deduct cash spent on equipment and working capital from operating cash flow to obtain FCF.

Illustrative classroom example. All figures and assumptions are hypothetical.

A parent approves a foreign affiliate's factory with NPV USD 3M. A local manager also guarantees USD 8M of a supplier's debt without central review.

- Assume a 25% probability of full payment on the guarantee next year, zero recovery, and a 10% risk-adjusted discount rate.
- The guarantee's expected payment is $0.25(8) = \text{USD } 2\text{M}$, with $\text{PV } 2/1.10 = \text{USD } 1.82\text{M}$.
- Net value after this additional obligation is $3 - 1.82 = \text{USD } 1.18\text{M}$. The board needs entity-level reporting and approval limits to see the exposure.

Interpretation

Governance affects value by changing actual obligations and investment choices.

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Illustrative classroom example. All figures and assumptions are hypothetical.

A controller owns 60% of Firm A and all of Firm B. Firm A sells an asset worth 10 to Firm B for 6. Use a common currency unit and ignore tax.

Owner / entity	Value effect
Firm A	-4.0
Controller through its stake in A	$0.60(-4) = -2.4$
Controller through B	+4.0
Controller net gain	+1.6
Minority shareholders of A	$0.40(-4) = -1.6$

Interpretation

The controller gains exactly what A's minority owners lose. Independent valuation and approval by disinterested shareholders can constrain this conflict.

Illustrative classroom example. All figures and assumptions are hypothetical.

A firm compares two one-year foreign-market strategies. All figures are USD M, FX is fixed, and the risk-adjusted required return is 10%.

Mode	Investment today	Year 1 parent cash	NPV
Export	4.0	5.0	$5/1.10 - 4 = 0.545$
Owned affiliate	8.0	10.0	$10/1.10 - 8 = 1.091$

A proposed guarantee adds USD 0.7M in present-value costs to the affiliate. Its revised NPV is $1.091 - 0.7 = 0.391$, below exporting.

Interpretation

Compare parent NPVs after investment, financing obligations, and governance costs for each entry mode.

Illustrative classroom example. All figures and assumptions are hypothetical.

A Hong Kong firm will receive USD 1M in 90 days. Dealer USD/HKD is 7.79 bid / 7.81 ask. The HKD interest rate is 4% simple, Actual/360.

- **Wrong side:** converting USD to HKD today would use 7.79, since the customer sells the base currency. The ask applies when buying USD.
- **Wrong horizon:** the 90-day interest factor is $1 + 0.04(90/360) = 1.01$.
- **Wrong timing:** today's bid applies to immediate delivery. Lock the future receivable with a forward or money-market hedge.
- **Units and intuition:** a lower HKD-per-USD rate means each dollar buys fewer HKD, reducing the unhedged receipt.

Interpretation

Check quote direction, trade side, interest horizon, and cash-flow date before calculating. State the economic meaning of the result.